

# Computer Science & IT

## Discrete Mathematics



**Set Theory & Algebra**

**Lecture No. 10**



**By- Vishal Sir**



# Recap of Previous Lecture



- ✓ Topic Lattice
- ✓ Topic Hasse diagram/POSET diagram
- ✓ Topic Bounded lattice
- ✓ Topic Complement of an element in a lattice
- ✓ Topic Complemented lattice



# Topics to be Covered



- ✓ **Topic** Complemented lattice
- ✓ **Topic** Sub-lattice
- ✓ **Topic** Distributive lattice
- ✓ **Topic** Boolean algebra



## Bounded Lattice

→ A lattice in which both minimum & maximum elements are present is called a bounded lattice.





## Topic : Bounded Lattice

Let  $[L, \vee, \wedge]$  be a lattice

① If there exists an element  $I \in L$

$$\text{s.t. } a \vee I = I \quad \forall a \in L$$

then 'I' is called Universal upper bound

(or) Upper bound of lattice L

"I" is actually the maximum element of lattice.

② If there exist an element  $O \in L$ .

$$\text{Such that, } a \wedge O = O, \quad \forall a \in L$$

then, 'O' is called Universal lower bound

(or) lower bound of lattice L

'O' is actually the minimum element

★ A lattice  $[L, \vee, \wedge]$  is called a bounded lattice if and only if both 'I' and 'O' exist in the lattice

(or)  
★ A lattice is called a bounded lattice if and only if both maximum & minimum element exist.



Note:- ① Every finite lattice is a bounded lattice

② If lattice is an infinite lattice then it may or may not be a bounded lattice

eg:  $(\mathbb{Z}, \leq)$  and  $(\mathbb{N}, \leq)$  are example of unbounded lattice  
Where both  $\mathbb{Z}$  &  $\mathbb{N}$  are infinite sets

Bounded lattice  $\rightarrow$  i.e. infinite set

eg: let  $A = \{x \mid x \in \mathbb{R} \text{ and } 0 \leq x \leq 1\}$   
 $\hookrightarrow A$  is an infinite set.

We know  $(A, \leq)$  is a lattice where  
'A' is an infinite set

but  $0 =$  Minimum element

&  $1 =$  Maximum element

$\rightarrow$  Both minimum & maximum element exist  
so it is a bounded lattice.





## Topic : Complement of an element in a lattice

\* Complements of an element of lattice  $[L, \vee, \wedge]$   
Can exist only if lattice 'L' is a bounded lattice

P only if Q

|||  
If P then Q

|||  
P  $\rightarrow$  Q

$\rightarrow$  (i) If Complement exist for an element of lattice  $[L, \vee, \wedge]$  then L is a bounded lattice

(ii) If Complement does not exist for an element of lattice  $[L, \vee, \wedge]$ , then 'L' may or may not be bounded

(iii) if  $[L, \vee, \wedge]$  is not a bounded lattice, then Complement can never exist for any element of lattice L





## Topic : Complement of an element in a lattice

Let  $[L, \vee, \wedge]$  be a bounded lattice.

Where,  $I$  = Universal upper bound of lattice  $L$  { i.e. Maximum element }  
&  $O$  = Universal lower bound of lattice  $L$  { i.e. Minimum element }

→ For an element  $a \in L$

if there exist some elements 'b' in set  $L$

Such that (i)  $a \vee b = I$  { Maximum }  $\Rightarrow$  i.e.  $\text{lub}(a, b) = \text{Maximum element}$   
& (ii)  $a \wedge b = O$  { Minimum }  $\Rightarrow$  i.e.  $\text{glb}(a, b) = \text{Minimum element}$

then,  $a$  &  $b$  are called Complement of each other w.r.t lattice  $[L, \vee, \wedge]$



## Note :

① In a bounded lattice,  
'I' and 'O' are always Complement of each other  
and they do not have any other Complement.

i.e., The only Complement of Maximum element (I) is minimum element (O)  
and The only Complement of minimum element (O) is maximum element (I)

## Imp.

② In a bounded lattice,  
Complement need not exist for every element  
of the lattice, and if exists it need not be unique  
*i.e., some elements may have more than one Complement*

→ In a bounded lattice Number of Complements of an element  
of the lattice may be '0' or '1' or more





## Topic : Complemented lattice

A lattice is called a Complemented lattice if and only if at least one Complement exist for Every element of the lattice

- In a Complemented lattice every element of the lattice will have 1 or more Complements



Today's Topic



Q:- Find the complement of Every element (if exist)  
of the following lattice

(i)  $(\{-1, 0, 2.5, 4, 6\}, \leq)$

$$\overline{0} = ?$$

$$\overline{2.5} = ?$$

$$\overline{4} = ?$$

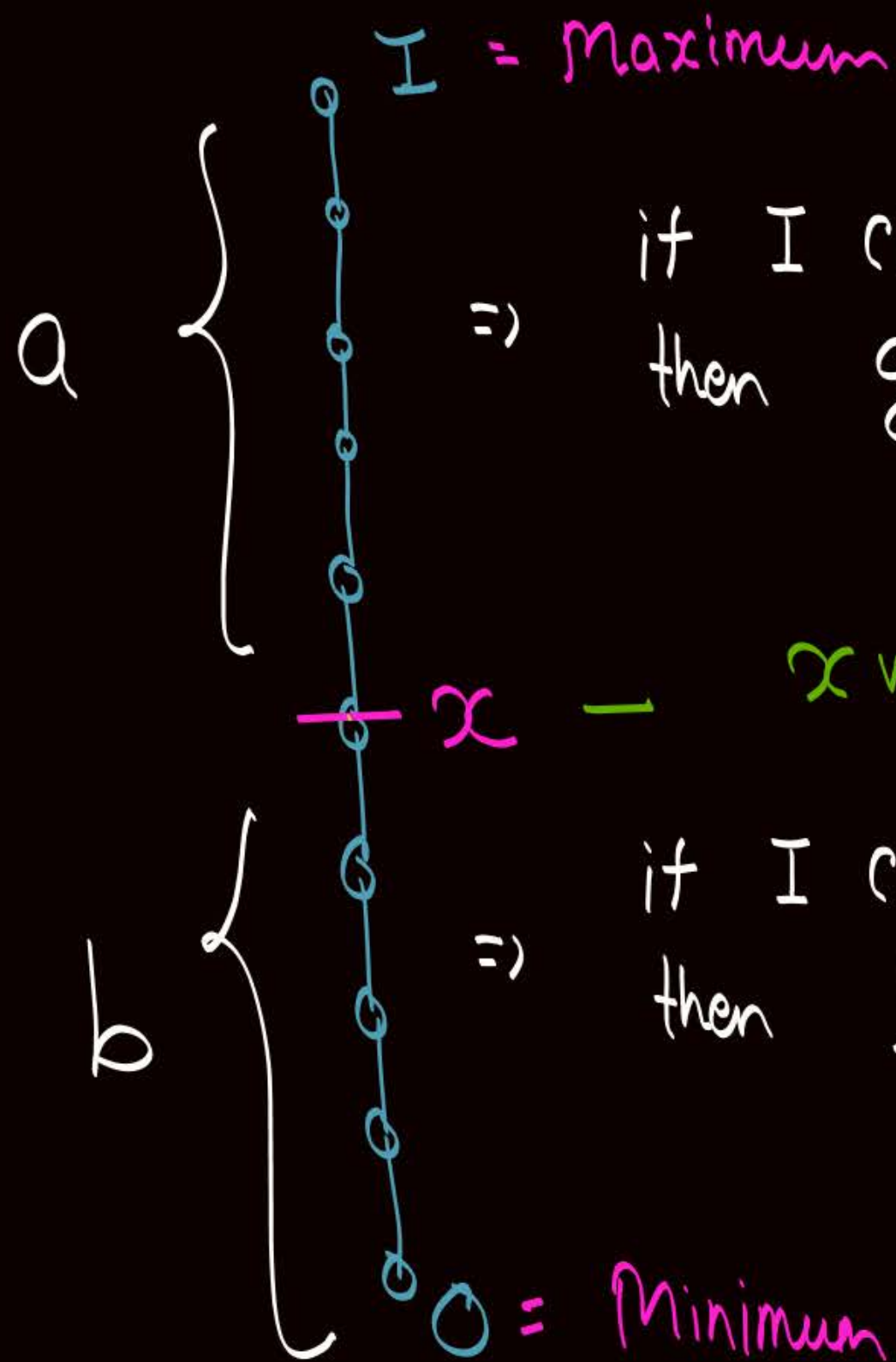


Maximum element = 6

Minimum element = -1

∴  $\boxed{\overline{-1} = 6 \text{ \& } \overline{6} = -1}$

they will not have  
any other complement



$\Rightarrow$  if I choose any element 'a' above 'x'  
 then  $\text{glb}(a, x) = x \neq \text{Minimum element}$

$$x \vee x = x \neq \text{Maximum} \therefore \bar{x} \neq x$$

$\Rightarrow$  if I choose any element 'b' below 'x'  
 then  $\text{lub}(x, b) = x \neq \text{Maximum element}$

$\therefore \bar{x}$  does not exist



Note:  $\rightarrow$  ① A TOSET is always a lattice, but  
Every lattice need not be a TOSET.

\* ② If TOSET is a bounded lattice, then  
Complement of minimum element = maximum element  
And Complement of maximum element = minimum element  
but Complement does not exist for any other  
element in a lattice w.r.t. TOSET

③ If two elements lie on the same line in a  
Hasse diagram, then they can not be complement  
of each other, Except for minimum & maximum element



Q:- Find the complement of Every element (if exist)  
of the following lattice

(i)  $(\{-1, 0, 2.5, 4, 6\}, \leq)$



Maximum element = 6

Minimum element = -1

∴  $\overline{-1} = 6 \text{ \& } \overline{6} = -1$

they will not have  
any other complement

Not a  
Complemented lattice

It is a TOSET,

∴  $\overline{0}$  = does not exist

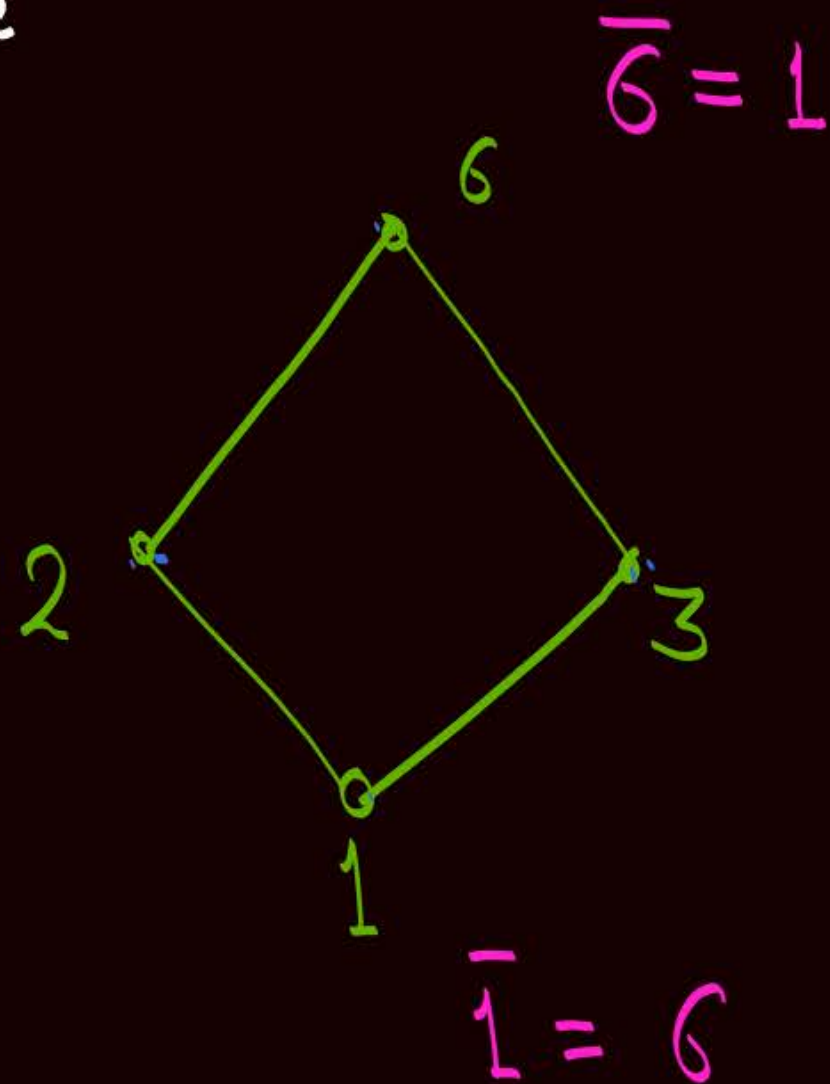
$\overline{2.5}$  = does not exist

$\overline{4}$  = does not exist



Q. Find the complement of Every element (if exist)  
 of the following lattice  
 (ii)  $(D_6, \div)$

$$\left. \begin{array}{l} \overline{4} = 6 \\ \overline{2} = 3 \\ \overline{3} = 2 \\ \overline{6} = 1 \end{array} \right\} \therefore \text{It is a Complemented lattice}$$



$$\begin{aligned} \text{lub}(2, 3) &= 6 = \text{Maximum} \\ \text{glb}(2, 3) &= 1 = \text{Minimum} \end{aligned}$$

$\therefore 2$  &  $3$  are  
Complement of  
each other

Q: Find the Complement of Every element (if exist)  
of the following lattice

(iii)  $(D_{12}, \div)$

$$\bar{1} = 12$$

$$\bar{12} = 1$$

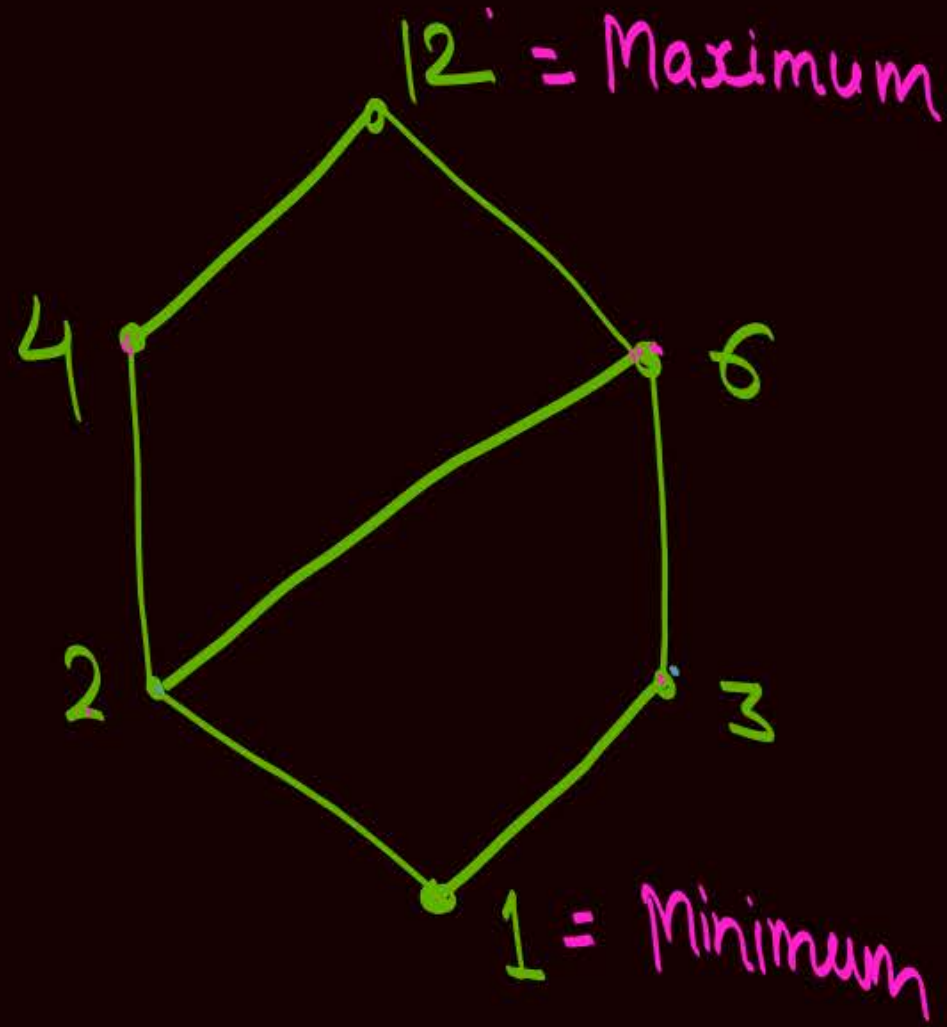
$$\bar{2} = \text{D.N.E.}$$

$$\bar{3} = 4$$

$$\bar{4} = 3$$

$$\bar{6} = \text{D.N.E.}$$

∴ Not a  
Complemented  
lattice



$\text{glb}(4, 6) = 2 \neq \text{Minimum}$   
∴ 4 & 6 can not  
be complement  
of each other

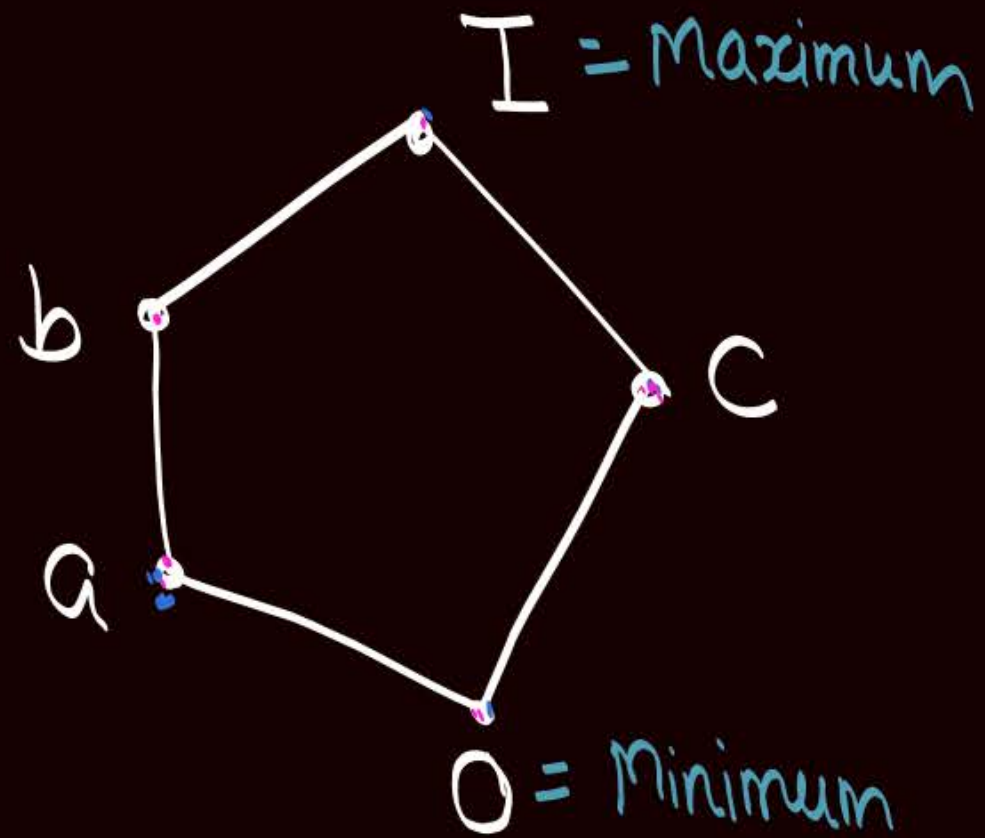
$\text{lub}(4, 3) = 12 = \text{Maximum}$

$\text{glb}(4, 3) = 1 = \text{Minimum}$

∴ 4 & 3 are  
Complement of  
each other

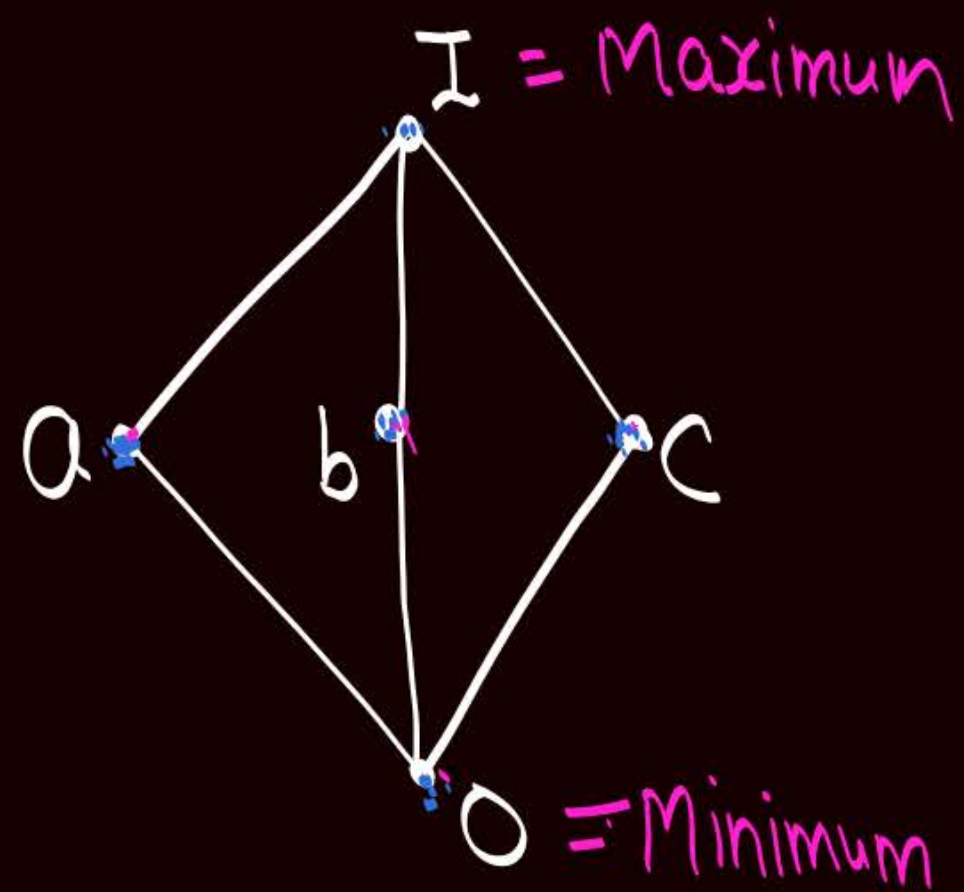


eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



$$\begin{array}{lcl}
 \overline{O} & = & I \\
 \overline{I} & = & O \\
 \overline{a} & = & c \\
 \overline{b} & = & c \\
 \overline{c} & = & a, b
 \end{array}
 \left. \vphantom{\begin{array}{lcl}} \right\} \text{is Complemented lattice}$$

eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



$$\overline{O} = I$$

$$\overline{I} = O$$

$$\overline{a} = b, c$$

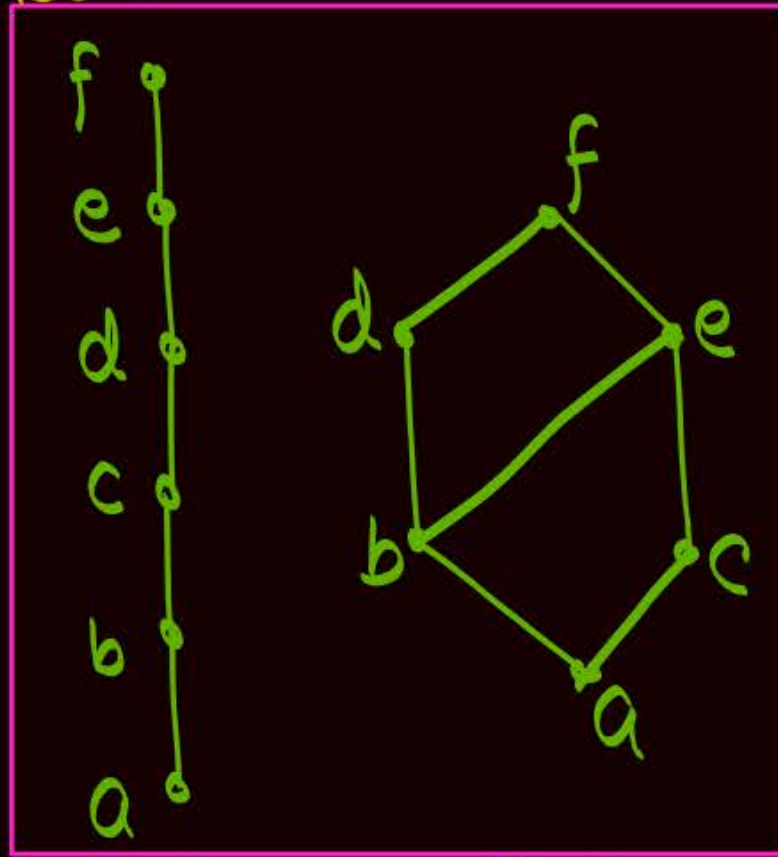
$$\overline{b} = a, c$$

$$\overline{c} = a, b$$

∴ Complemented lattice



Note:-



$$\bar{a} = f$$

$$\bar{f} = a$$

$$\bar{a} = f$$

$$\bar{f} = a$$

$$\bar{c} = d$$

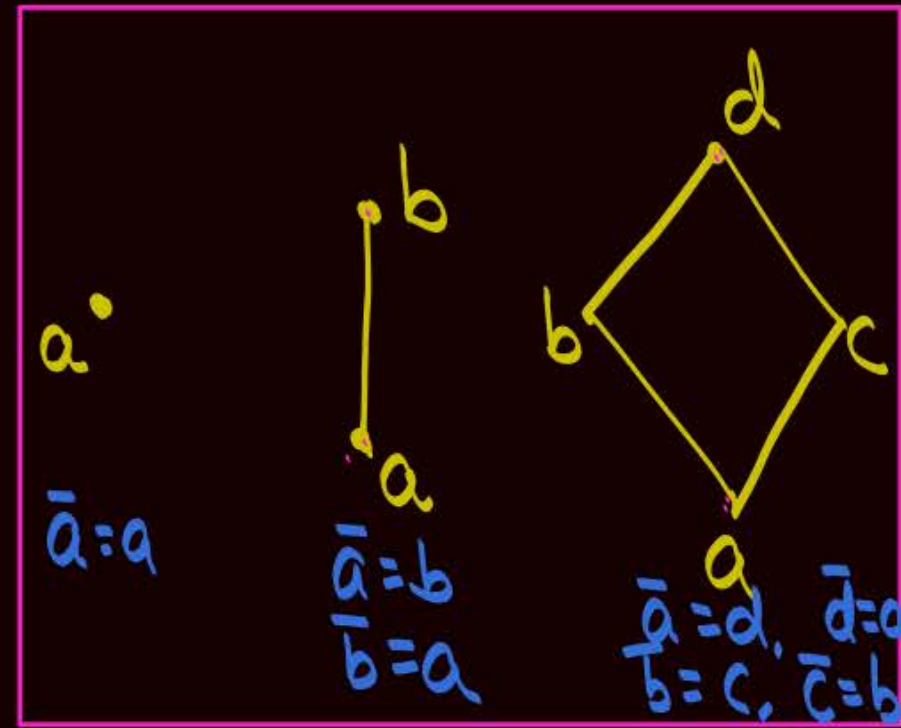
$$\bar{d} = c$$

$$\bar{b} = \text{DNE}$$

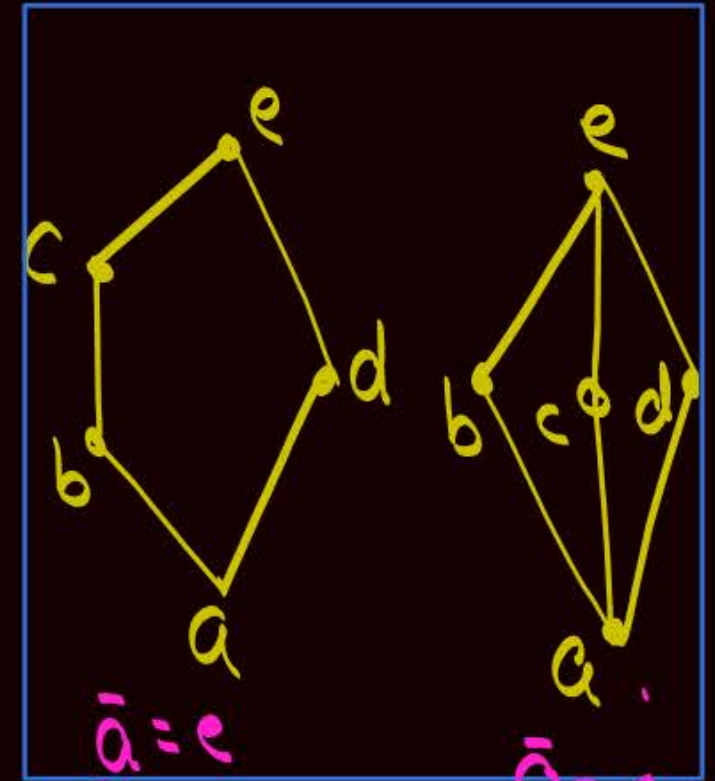
$$\bar{e} = \text{DNE}$$

Complement does not exist for any other element

These are not complemented lattice



Every element has Exactly one Complement



$$\bar{a} = e$$

$$\bar{e} = a$$

$$\bar{b} = d$$

$$\bar{c} = d$$

$$\bar{d} = b, c$$

$$\bar{a} = e$$

$$\bar{e} = a$$

$$\bar{b} = c, d$$

$$\bar{c} = b, d$$

$$\bar{d} = b, c$$

All are complemented lattice





## Topic : Sublattice

→ Let  $[L, \vee, \wedge]$  be a lattice, and  $M$  is a subset of set  $L$

$M$  is a sub-lattice of lattice  $L$  if and only if

(i)  $[M, \vee, \wedge]$  must be a lattice

{ i.e. lub and glb for every pair of element of set  $M$ , must exist in set  $M$  itself }

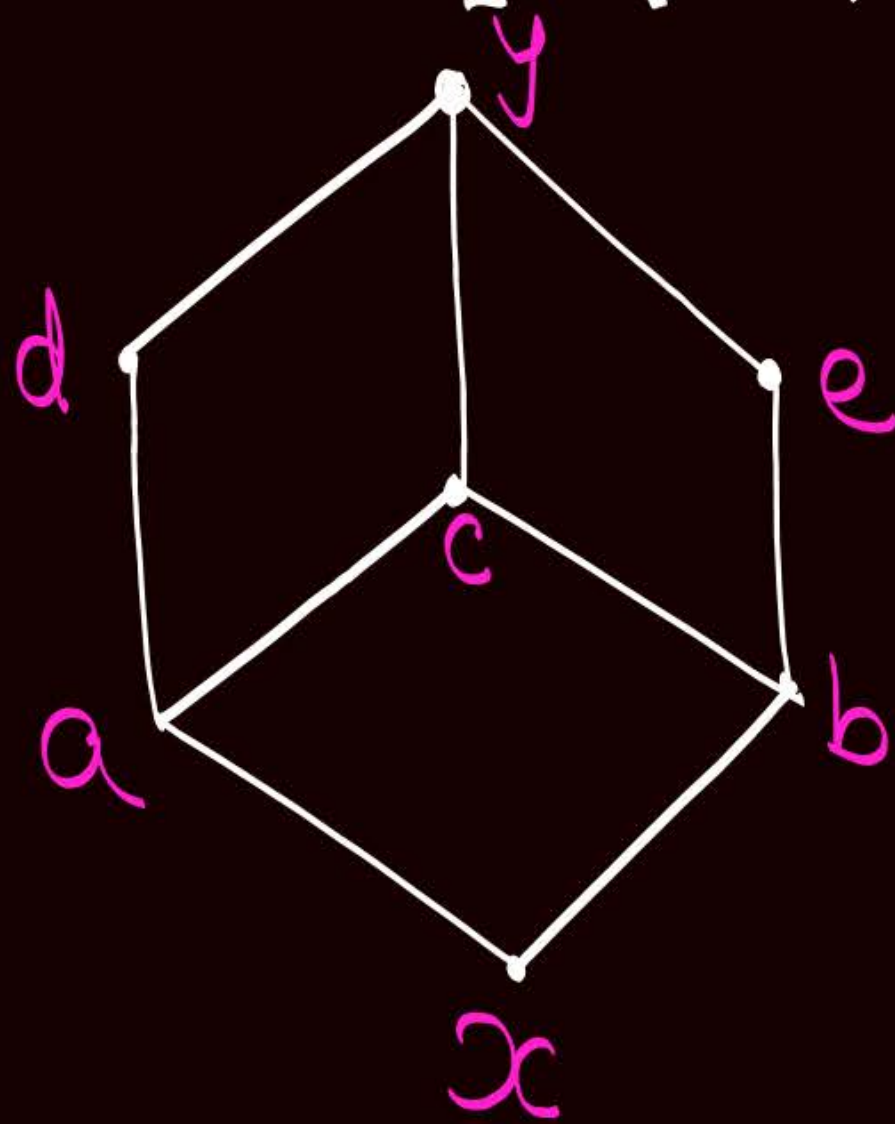
and (ii) lub and glb for every pair of elements in lattice  $[M, \vee, \wedge]$  must be exactly same as lub and glb for that pair of element in lattice  $[L, \vee, \wedge]$

i.e,  $\forall a, b \in M$   
lub  $(a, b)$  in  $M$  = lub  $(a, b)$  in  $L$   
glb  $(a, b)$  in  $M$  = glb  $(a, b)$  in  $L$



eg Consider the following Lattice 'L'

"L" = {x, y, a, b, c, d, e}



Which of the following is/are sublattice of lattice L

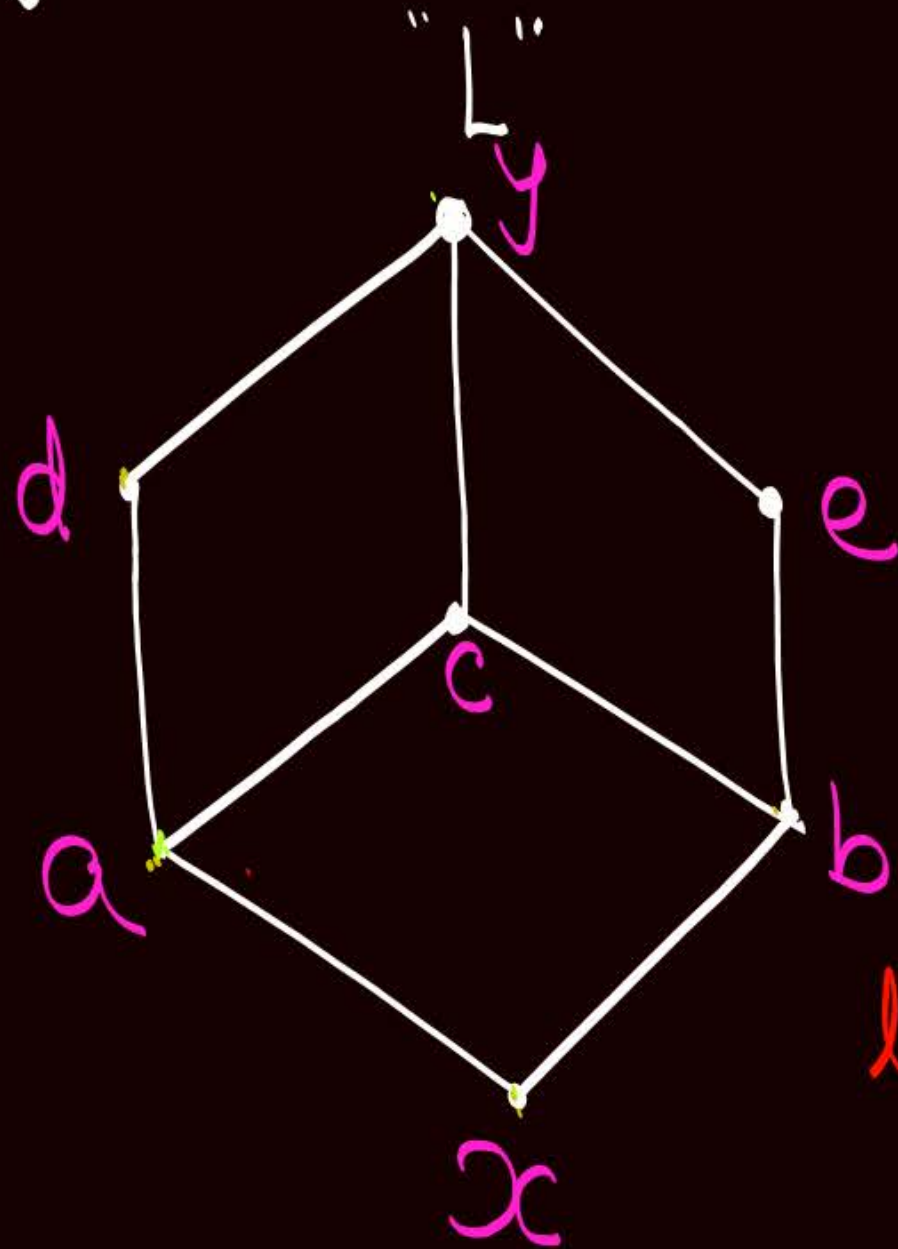
(i)  $L_1 = \{x, a, b, y\}$

(ii)  $L_2 = \{x, c, d, y\}$

(iii)  $L_3 = \{x, a, c, y\}$

(iv)  $L_4 = \{x, a, e, y\}$

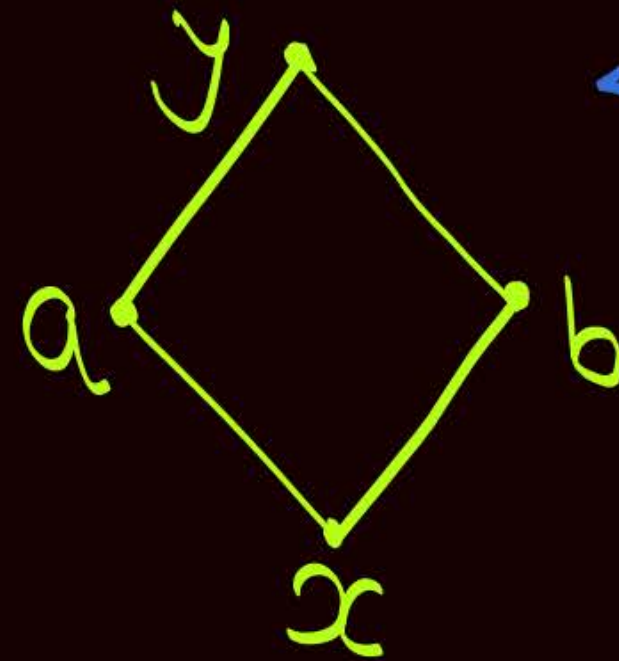
eg Consider the following lattice 'L'



$$\text{lub}(a, b) = c$$

Which of the following is/are sublattices of lattice L

(i)  $L_1 = \{x, a, b, y\}$



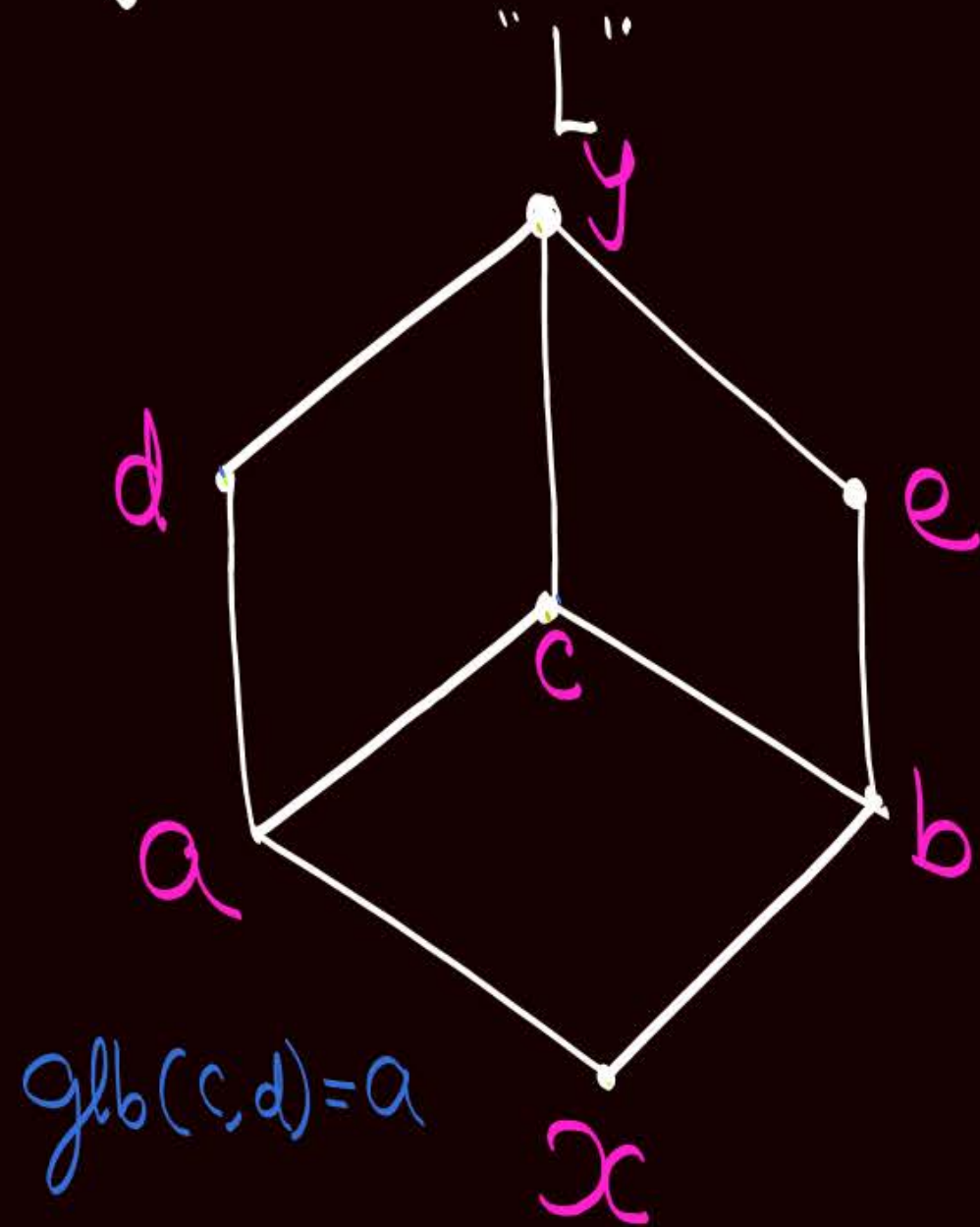
← It is a lattice

$$\text{lub}(a, b) = y \neq \text{lub}(a, b) \text{ in } L$$

∴  $L_1$  is not a sub-lattice of L



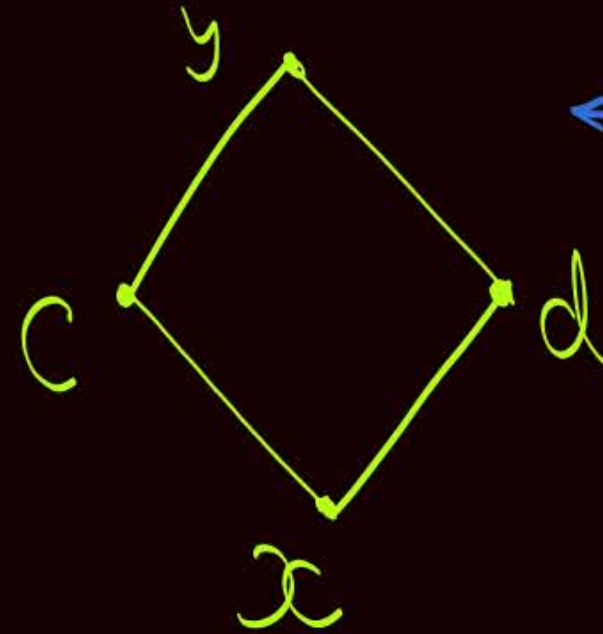
eg Consider the following Lattice 'L'



$$\text{glb}(c, d) = a$$

Which of the following is/are sublattices of lattice L

(ii)  $L_2 = \{x, c, d, y\}$

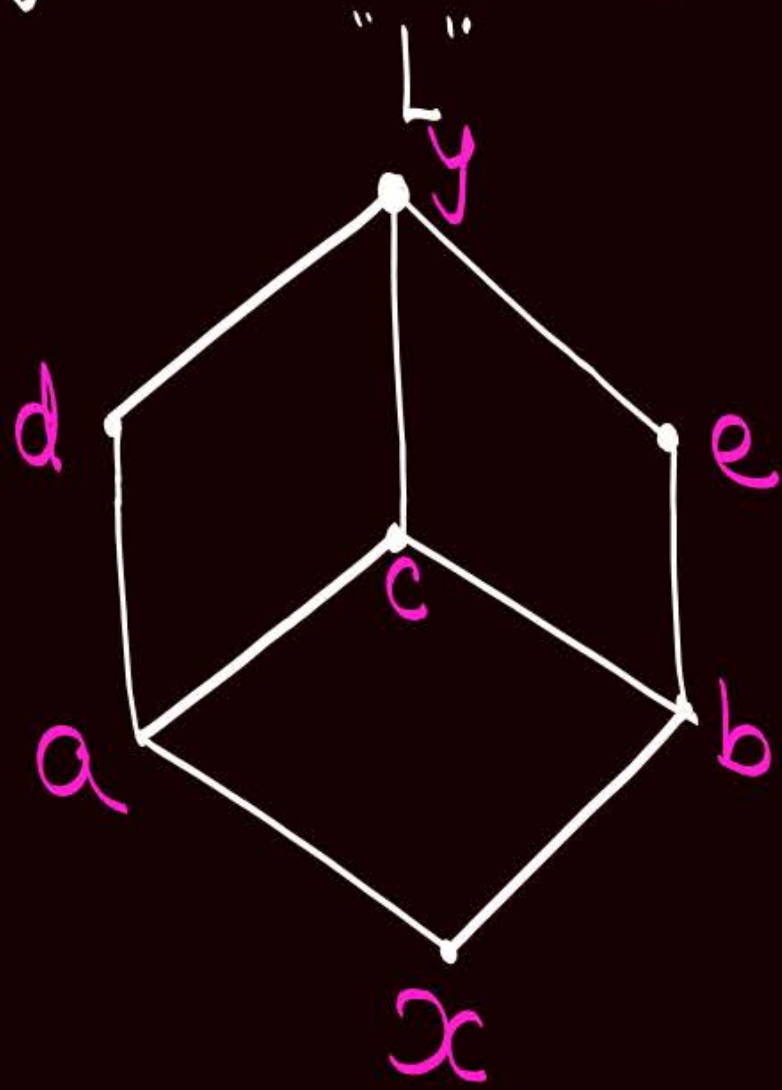


← it is a lattice

$$\text{glb}_{\text{in } L_2}(c, d) = x \neq \text{glb}_{\text{in } L}(c, d)$$

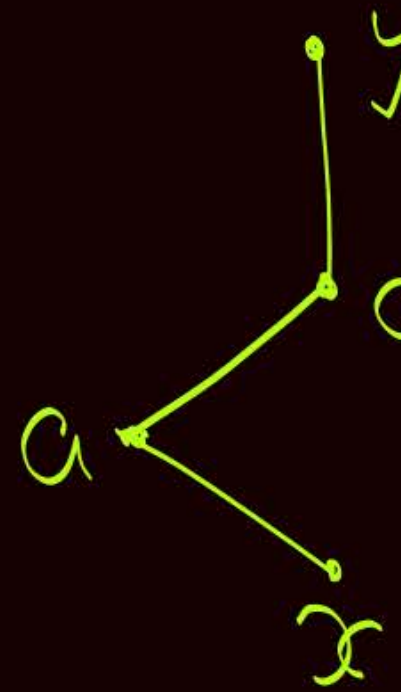
∴ L<sub>2</sub> is not a sub-lattice of L

eg Consider the following Lattice 'L'



Which of the following is/are  
Sublattices of lattice L

(iii)  $L_3 = \{x, a, c, y\}$



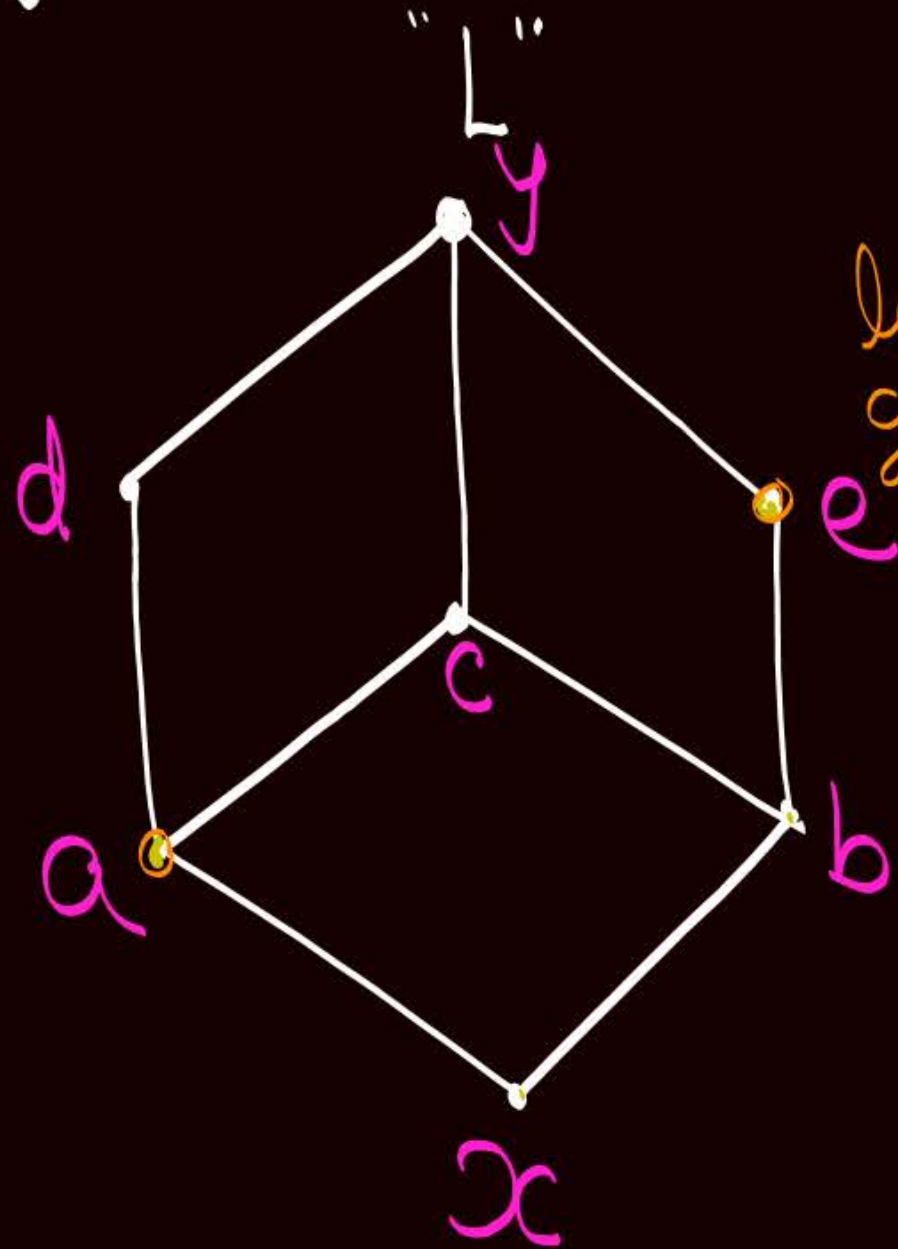
← It is a lattice

← lub & glb for every  
Pair of elements in  $L_3$   
is same as lub & glb  
for that pair of element  
in lattice L

∴  $L_3$  is a  
Sub-lattice of L



eg Consider the following Lattice 'L'

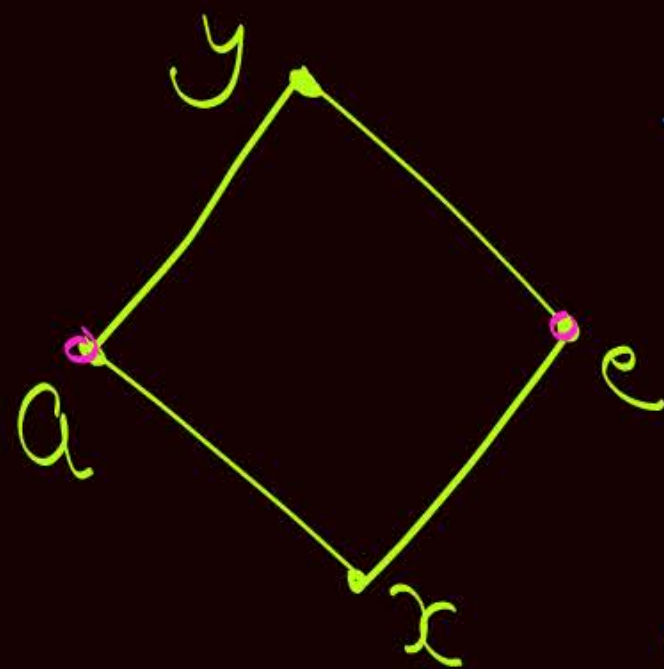


$$\begin{aligned} \text{lub}(a, e) &= y \\ \text{glb}(a, e) &= x \end{aligned}$$

$$\begin{aligned} \text{lub}(a, e) &= y \\ \text{glb}(a, e) &= x \end{aligned}$$

Which of the following is/are sublattices of lattice L

(iv)  $L_4 = \{x, a, e, y\}$



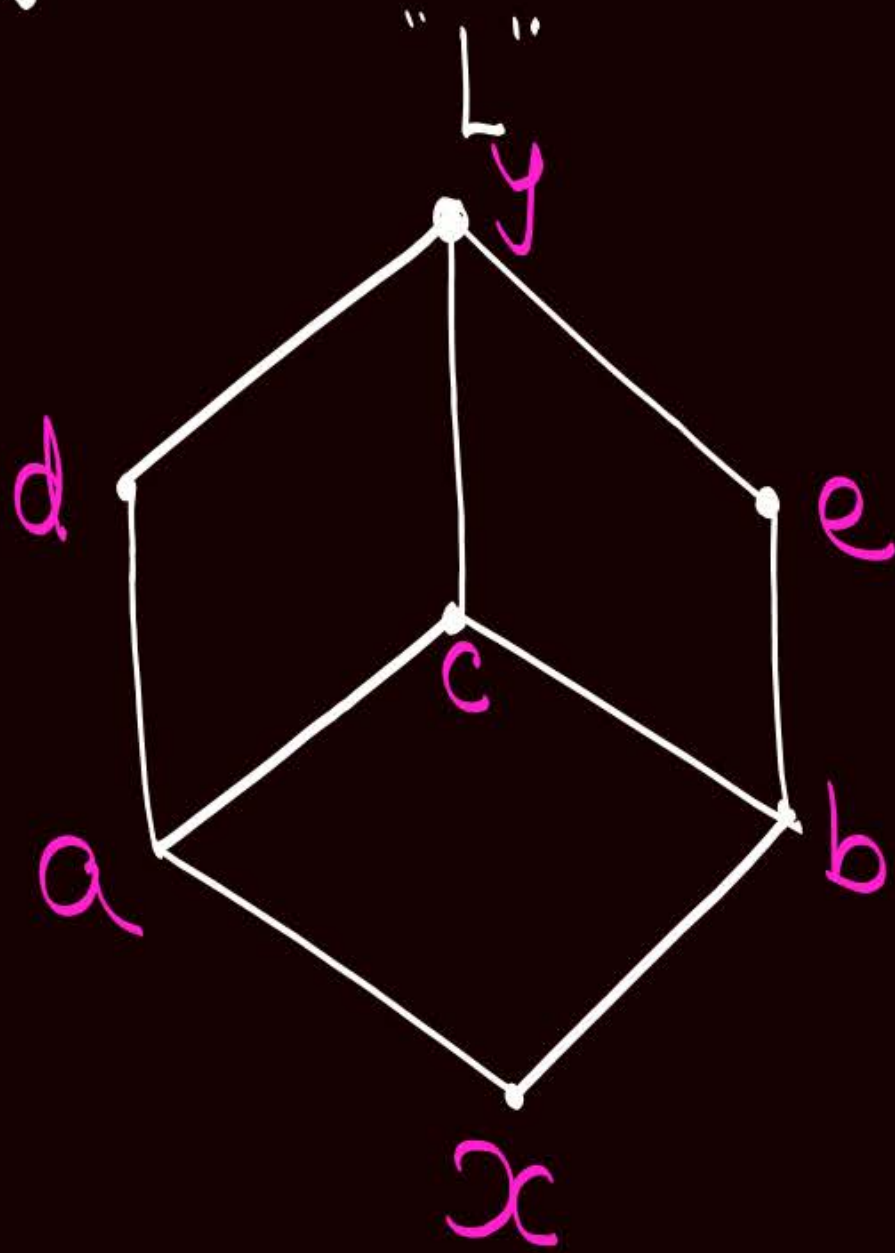
← it is a lattice

$$\text{lub}(a, e)_{\text{in } L_4} = y = \text{lub}(a, e)_{\text{in } L}$$

$$\text{glb}(a, e)_{\text{in } L_4} = x = \text{glb}(a, e)_{\text{in } L}$$

∴  $L_4$  is a Sub-lattice of L

eg Consider the following Lattice 'L'



Which of the following is/are sublattices of lattice L

(i)  $L_1 = \{x, a, b, y\}$

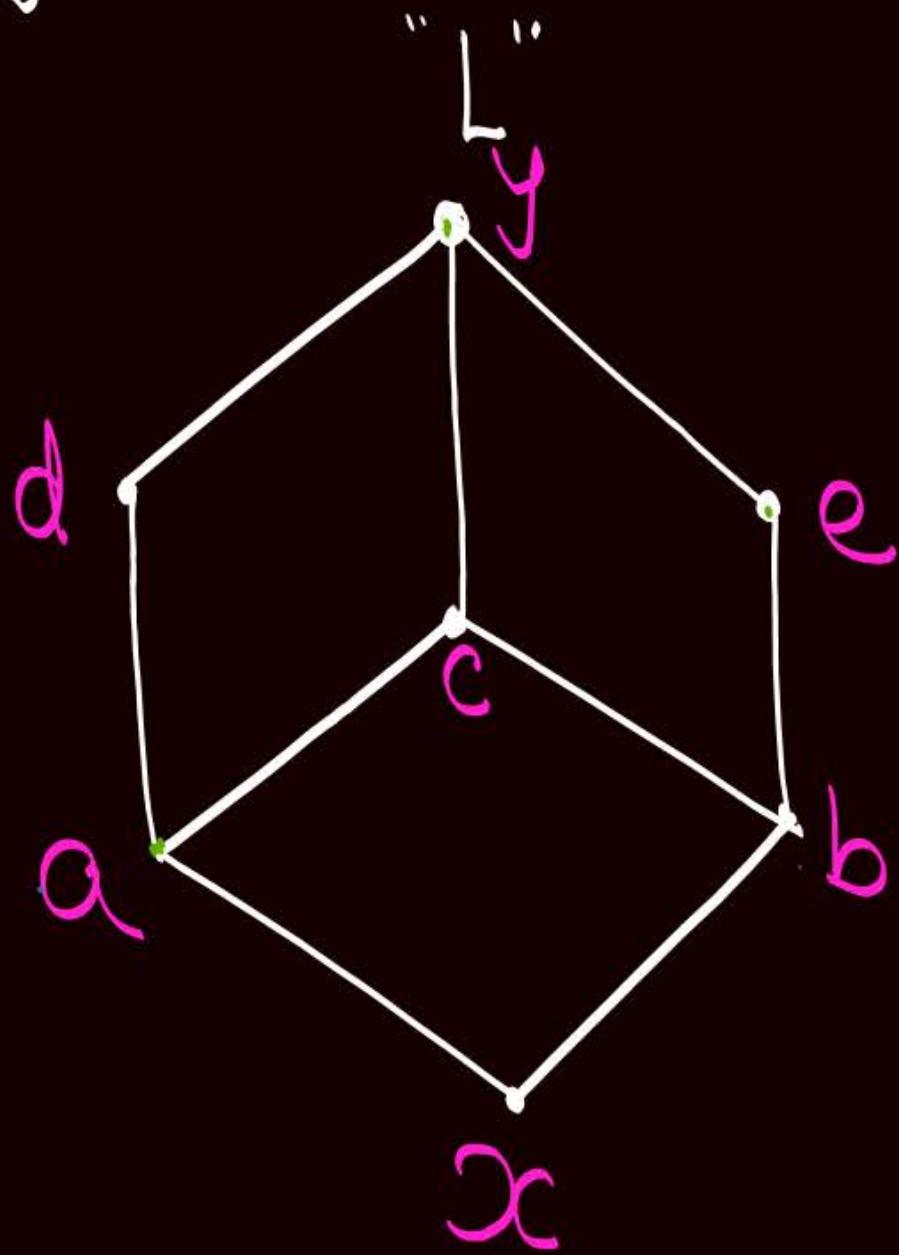
(ii)  $L_2 = \{x, c, d, y\}$

~~(iii)~~  $L_3 = \{x, a, c, y\}$

~~(iv)~~  $L_4 = \{x, a, e, y\}$



eg Consider the following Lattice 'L'



Which of the following is/are  
sub-lattices of lattice L

(a)  $L_1 = \{a, e, y\}$



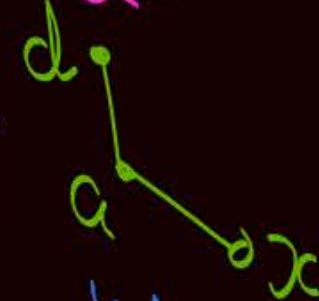
← it is not a lattice

(b)  $L_2 = \{a, b, c\}$



← it is not a lattice

(c)  $L_3 = \{x, a, d\}$



→ it is a sub-lattice of L

(d)  $L_4 = \{a, c, y\}$



it is a sub-lattice of L



## Topic : Distributive lattice

Let  $[L, \vee, \wedge]$  be a lattice.

$[L, \vee, \wedge]$  is a distributive lattice if and only if,

✓(i)  $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

And ✓(ii)  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \leftarrow \forall a, b, c \in L$

Lattice is called a distributive lattice if and only if distributive property is satisfied by Every combination of elements of the Lattice



Note:-

① In a distributive lattice every element has at most one complement

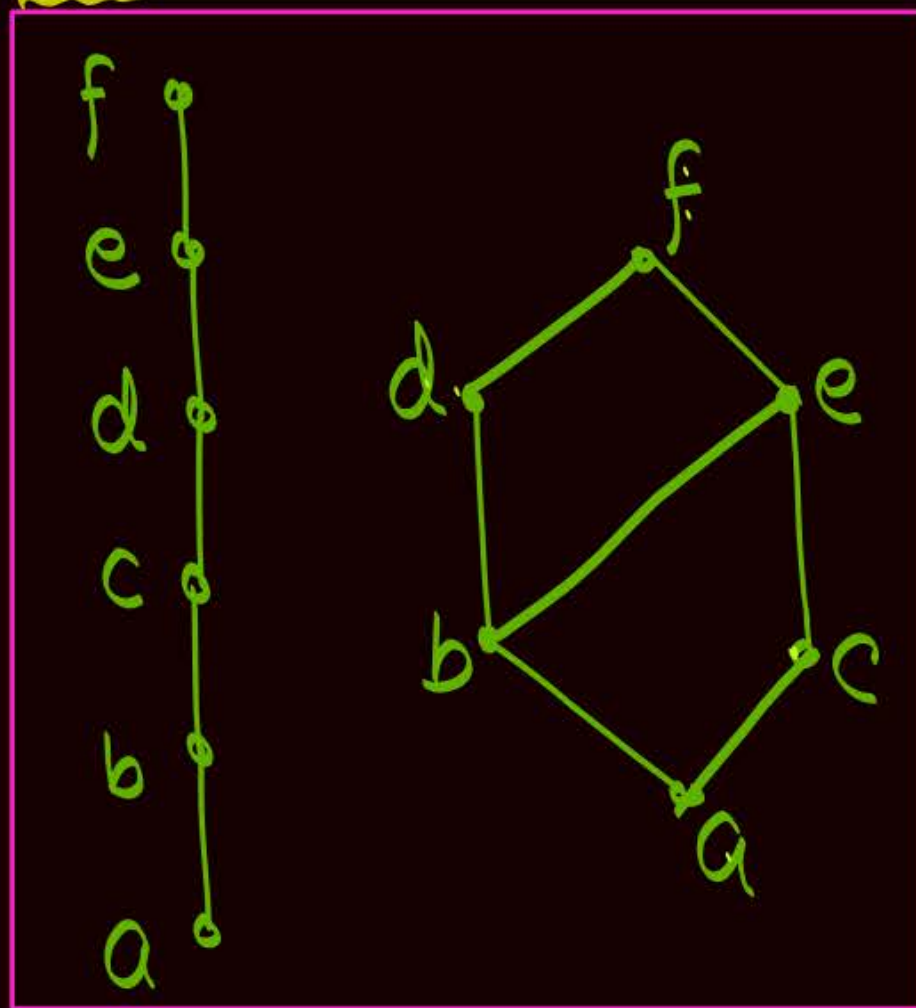
If lattice is a <sup>|||</sup>distributive lattice, then every element will have at most one complement, but converse of the statement need not be true

Note:-

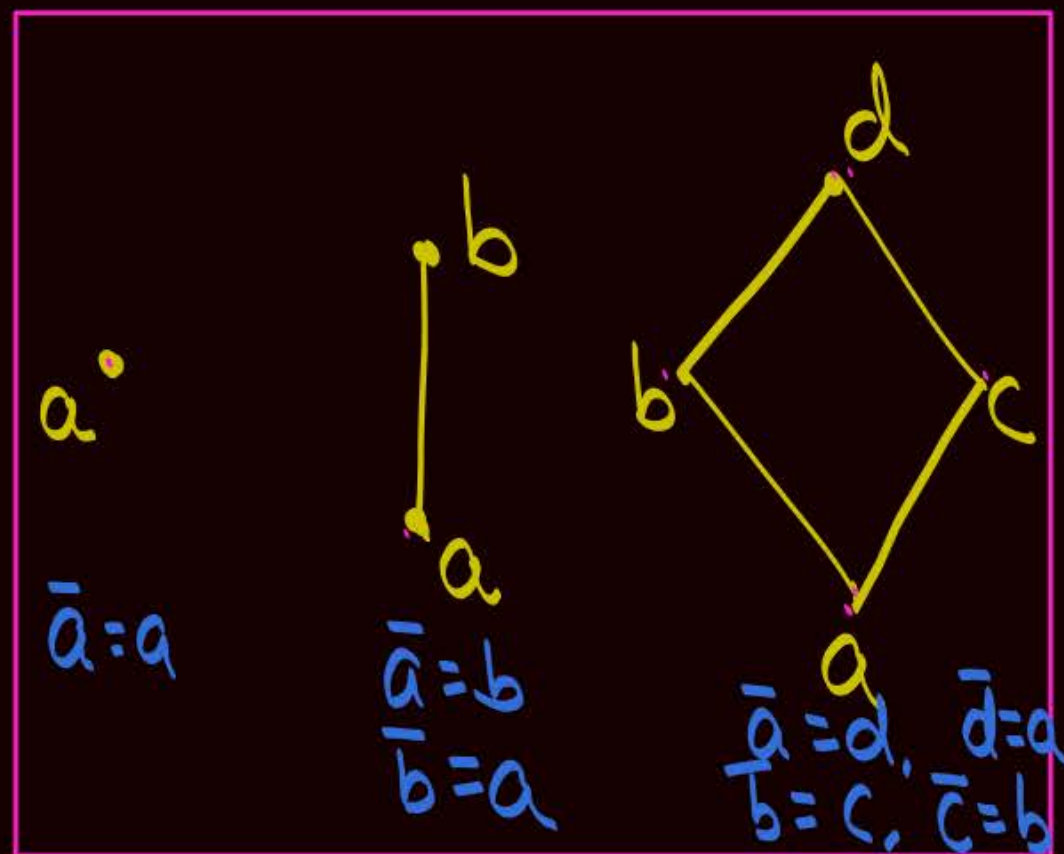
- ② In a lattice, If any element of the lattice has more than one Complement, then that lattice is not a distributive lattice
- ③ In a lattice, if Every element has exactly '1' Complement, then that lattice is definitely a distributive lattice
- ④ In a lattice, if Every element has at most one Complement, 'but not Exactly one', then that lattice may or may not be distributive



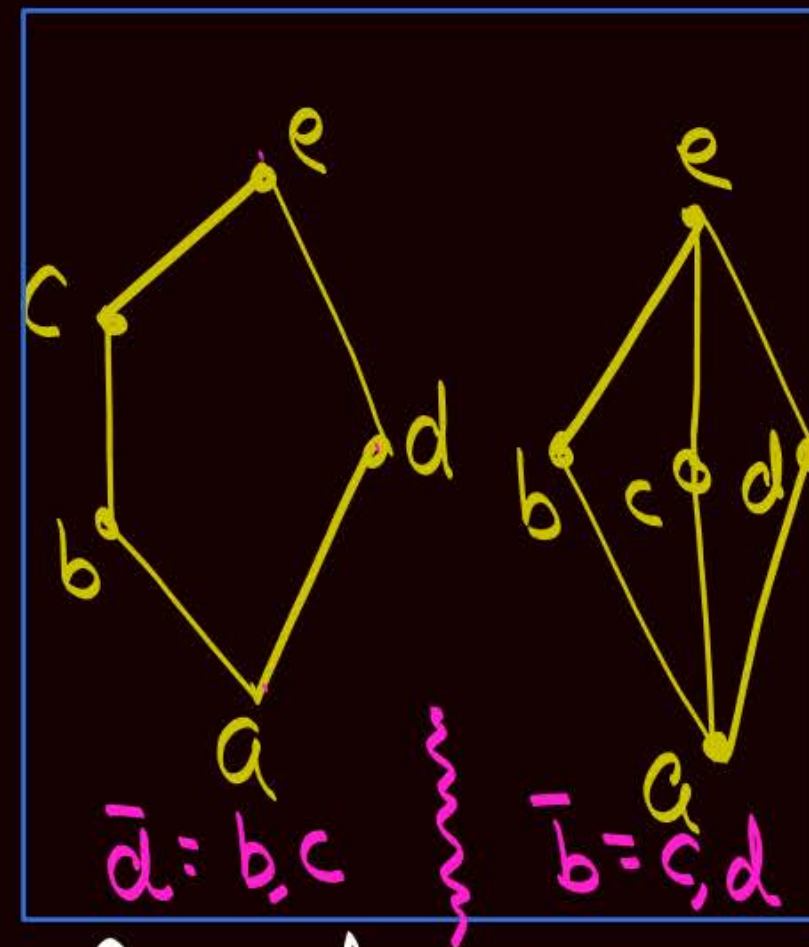
Note:-



- Examples in which Every element has '0' or '1' Complement
- or We need to check for distributive property



- Every element has Exactly one Complement.
- i. Above lattice are distributive lattice



- Some elements have More than One Complement
- ∴ Above lattice are not distributive lattice

Note :  $\rightarrow$  ① If we choose elements from the same line of "Hasse diagram", then distributive property will always be satisfied by those elements.



② If Lattice is a TOSET, then distributive property will always be satisfied by the elements of that lattice

$\therefore$  A TOSET is always a distributive lattice



Note:

There '4' TOSET  
are w.r.t. distributive

a •

Hasse diagram  
w.r.t lattice  
of size = 1



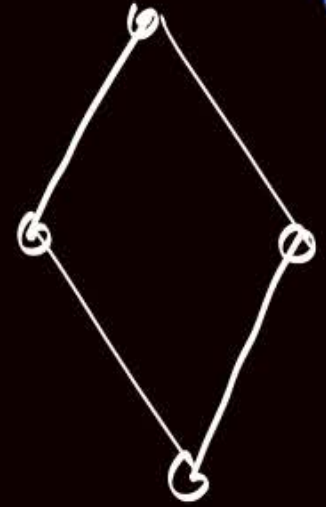
Hasse diagram  
w.r.t lattice  
of size = 2



Hasse diagram  
w.r.t lattice  
of size = 3



Hasse diagrams  
w.r.t lattice  
of size = 4



Every element  
has Exactly  
One Complement  
∴ distributive

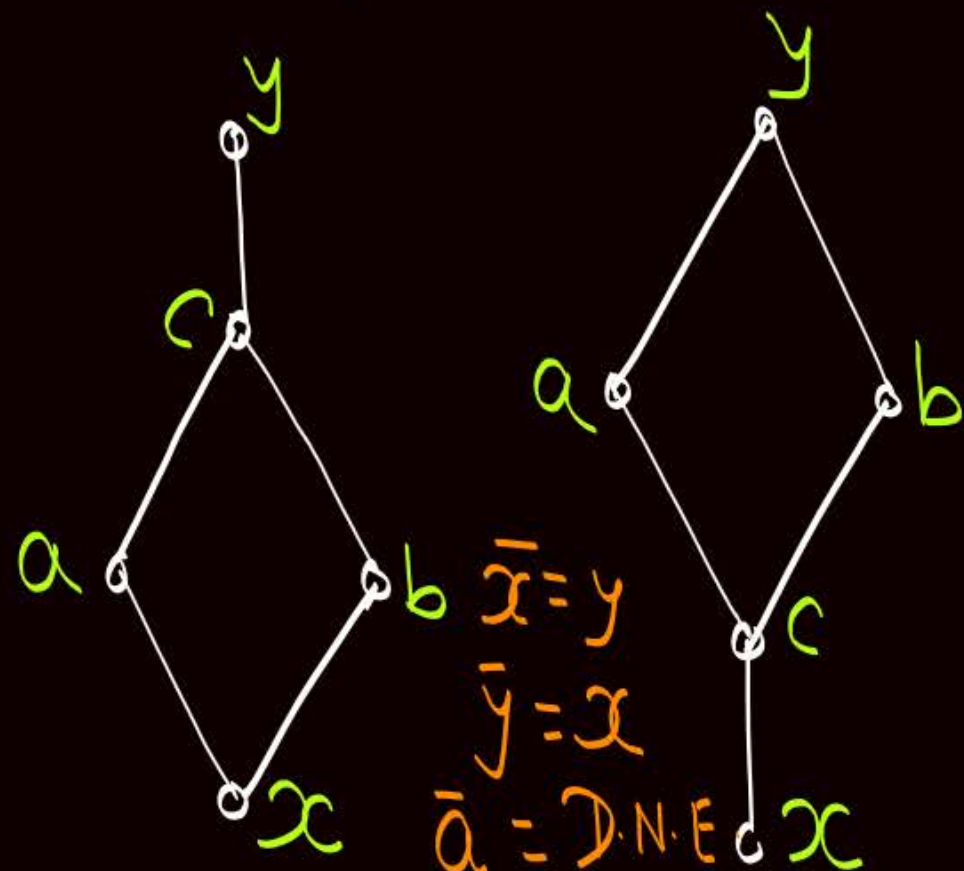
Note: Any lattice with '4' or less elements is always distributive

Note:

Hasse diagrams possible w.r.t. lattice of size = 5

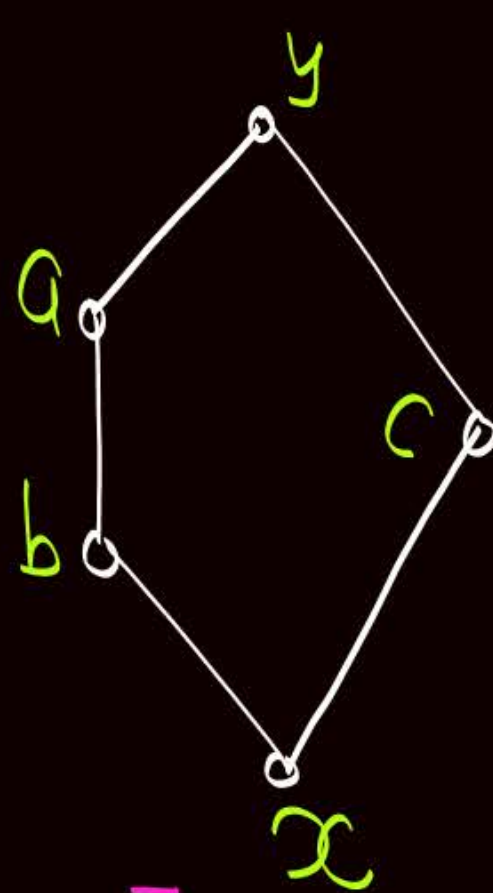


w.r.t. TOSET  
is Distributive

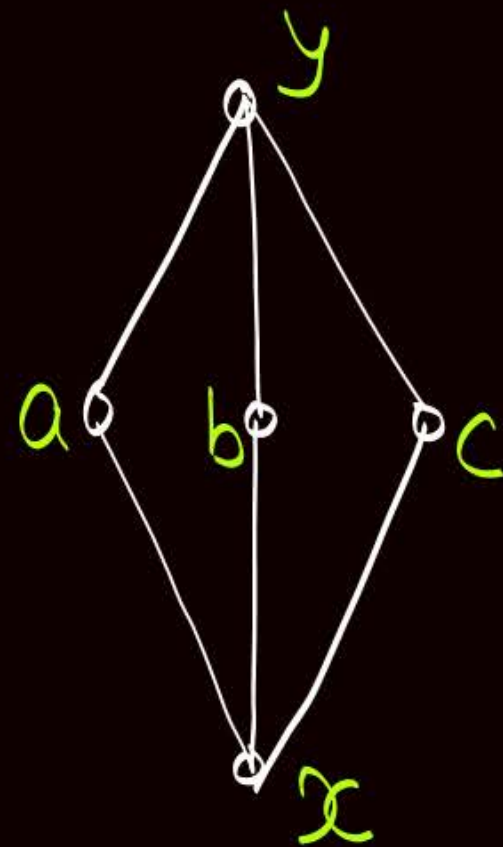


$\bar{x} = y$   
 $\bar{y} = x$   
 $\bar{a} = \text{D.N.E.}$   
 $\bar{b} = \text{D.N.E.}$   
 $\bar{c} = \text{D.N.E.}$

Every element has at most  
One complement, but these  
are distributive



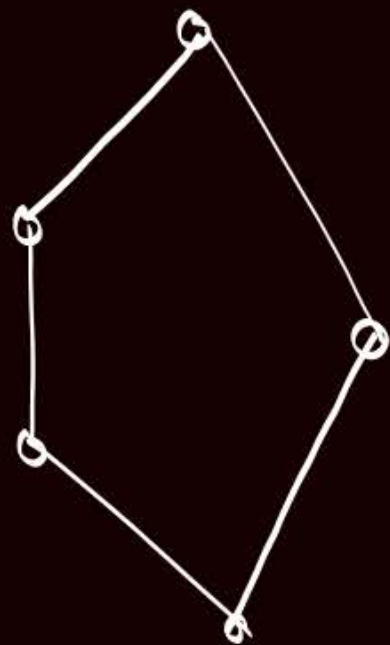
$\bar{c} = a, b$



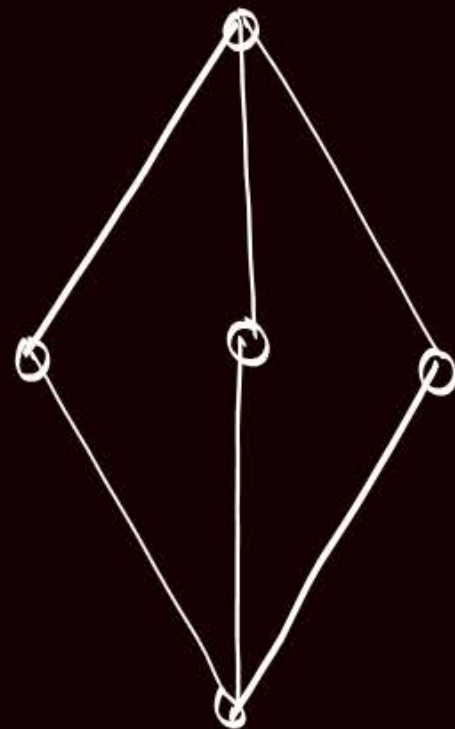
$\bar{a} = b, c$

Some elements have more than  
one complement is Not distributive





$L_1^*$



$L_2^*$

\*  $L_1^*$  and  $L_2^*$  are only two lattice of size = 5 that are not distributive

Note :-  $\rightarrow$

A lattice  $L$  is not distributive  
if and only if, there exist a  
sub-lattice of lattice  $L$  which  
is isomorphic to  $L_1^*$  or  $L_2^*$

important

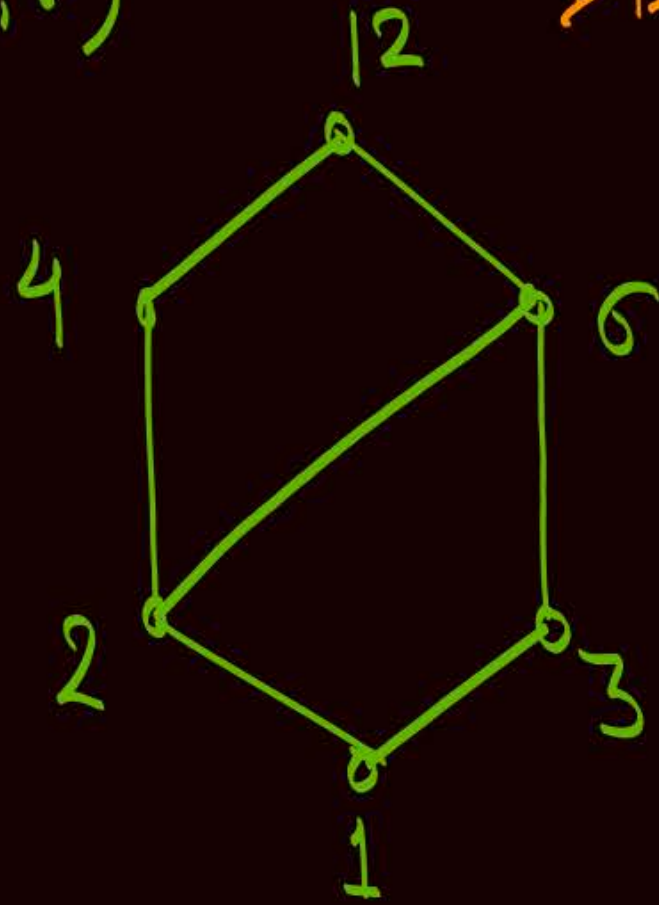
looks similar



Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$

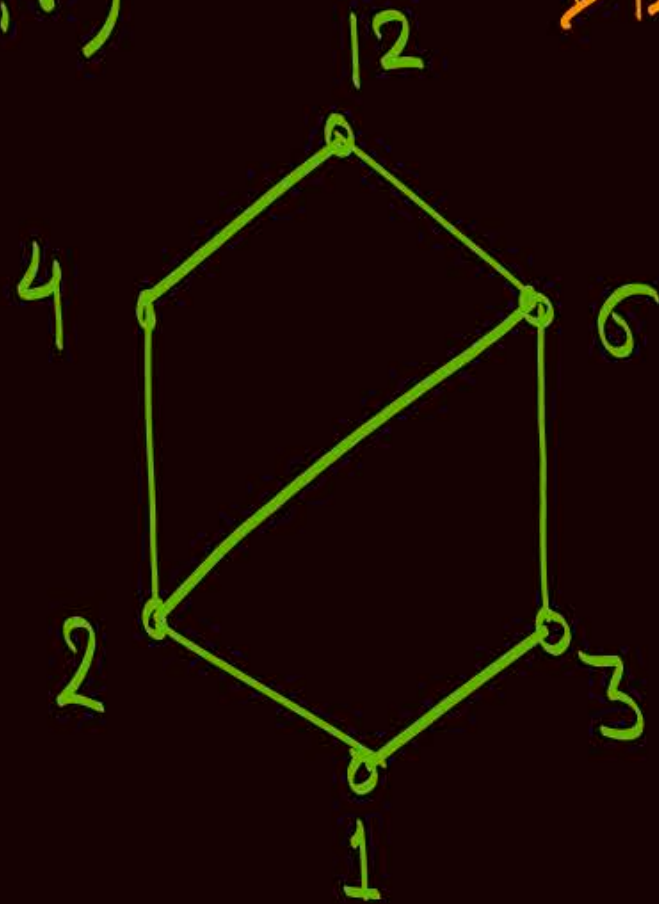
$D_{12} = \{1, 2, 3, 4, 6, 12\}$



Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$

$D_{12}: \{1, 2, 3, 4, 6, 12\}$



① No. of elements  $> 4$ ,  $\therefore$  May or May not be distributive

② Not a TOSET,  $\therefore$  May not be distributive

③  $\bar{1} = 6$   
 $\bar{12} = 1$   
 $\bar{2} = \text{DNE}$   
 $\bar{3} = 4$   
 $\bar{4} = 3$   
 $\bar{6} = \text{DNE}$

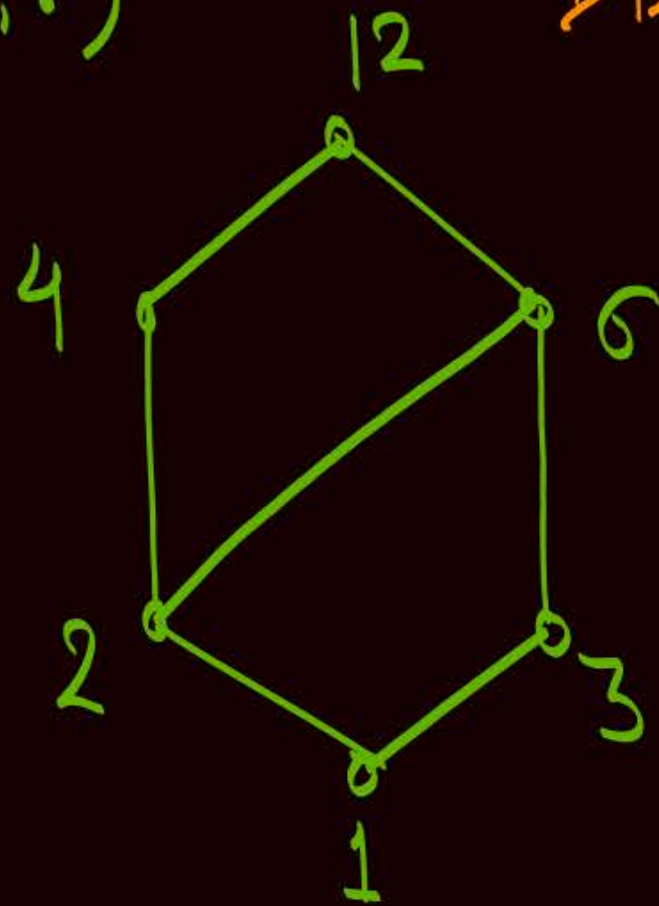
} Every element has at most '1' Complement,  $\therefore$  May or May not be distributive.



Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$

$D_{12}: \{1, 2, 3, 4, 6, 12\}$

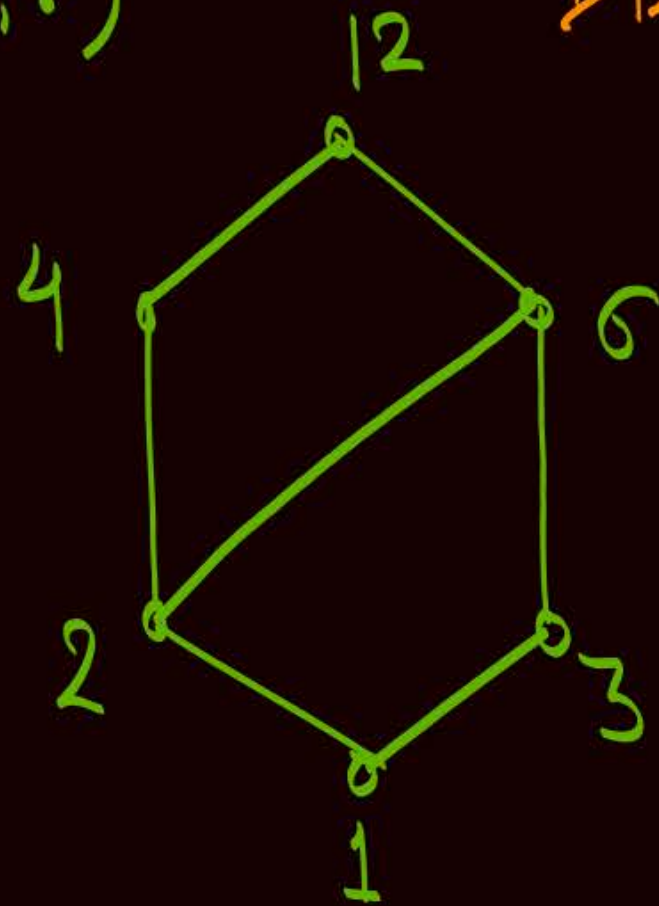


→ Do we have any subset of set  $\{1, 2, 3, 4, 6, 12\}$  s.t. the Hasse diagram is isomorphic to  $L_1^* \text{ or } L_2^*$  ✓

Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$



→ Do we have any subset of set  $\{1, 2, 3, 4, 6, 12\}$ , s.t. the Hasse diagram is isomorphic to  $L_1^*$  or  $L_2^*$

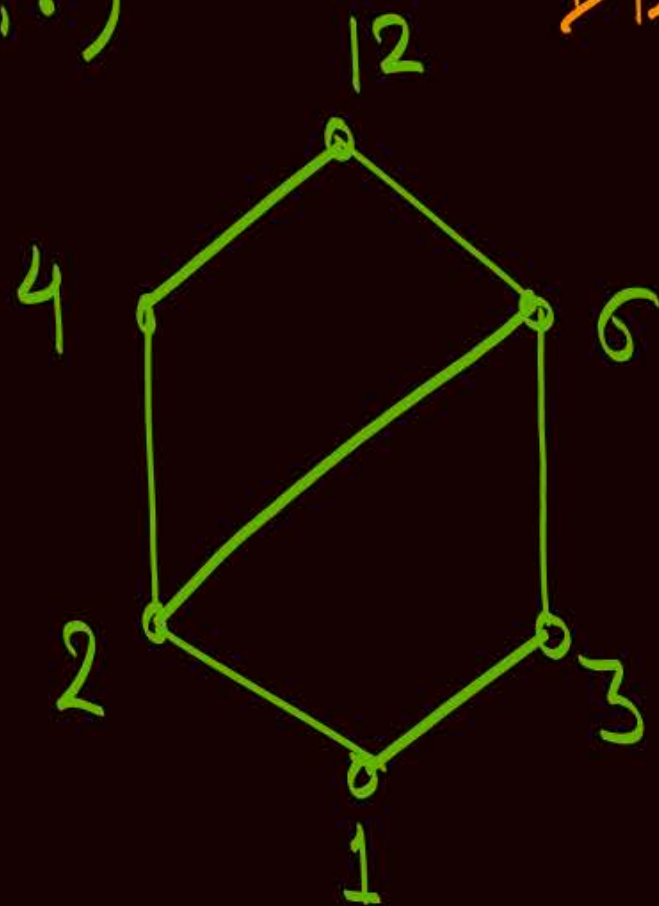
① → We can not choose any combination of '5' elements s.t. Hasse diagram is isomorphic to  $L_2^*$



Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$

$D_{12}: \{1, 2, 3, 4, 6, 12\}$



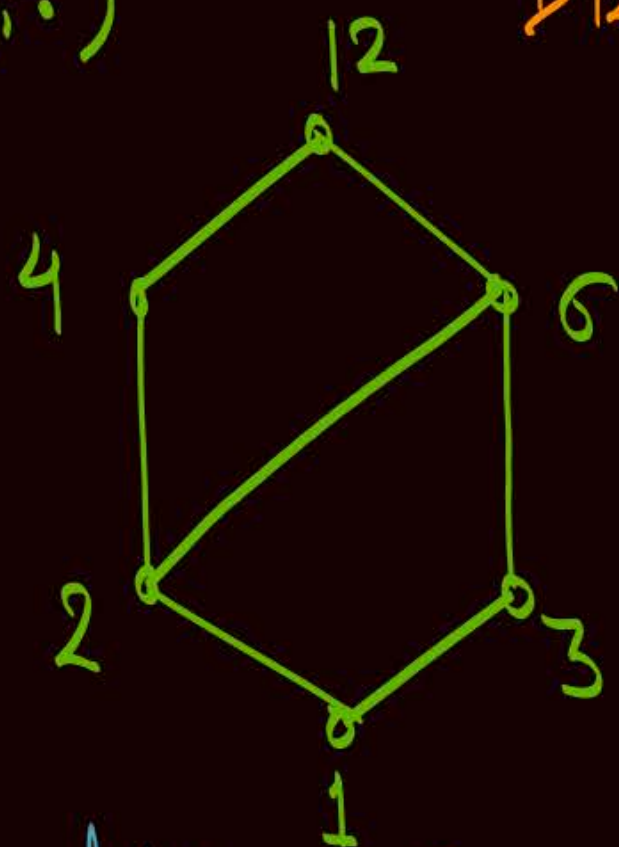
→ Do we have any subset of set  $\{1, 2, 3, 4, 6, 12\}$ , s.t. the Hasse diagram is isomorphic to  $L_1^*$  or  $L_2^*$

②  $\{1, 2, 3, 4, 12\}$  &  $\{1, 3, 4, 6, 12\}$  are two subsets s.t. Hasse diagram will be isomorphic to  $L_1^*$

Q:- Check whether the given lattice is distributive or not?

( $D_{12}, \div$ )

$D_{12}: \{1, 2, 3, 4, 6, 12\}$



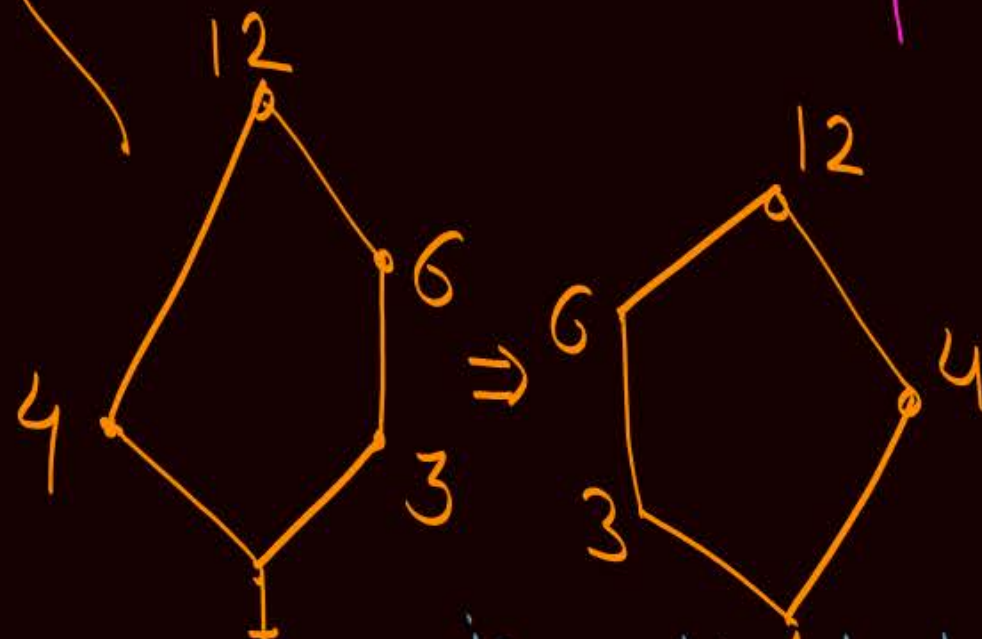
$$\text{lub}(2, 3) = 6$$

$$\text{glb}(4, 6) = 2$$

②  $\{1, 2, 3, 4, 12\}$  &  $\{1, 3, 4, 6, 12\}$  are two subsets s.t. Hasse diagram will be isomorphic to  $L_1^*$



isomorphic to  $L_1^*$   
but  $\text{lub}(2, 3) = 12 \neq \text{given lattice}$   
Hence  $\{1, 2, 3, 4, 12\}$  is not a sub-lattice



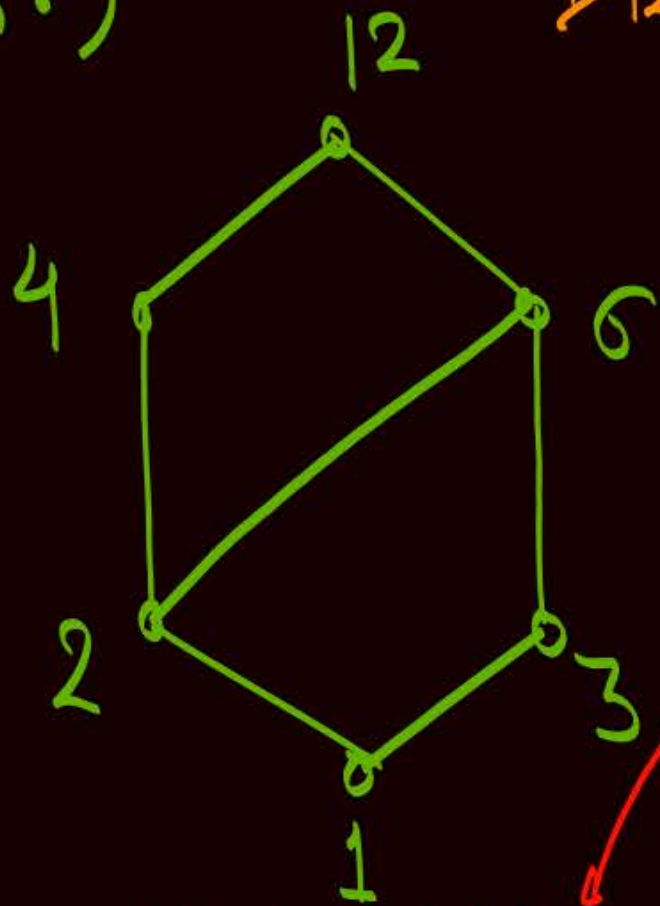
isomorphic to  $L_1^*$   
but  $\text{glb}(4, 6) = 1 \neq \text{given lattice}$   
Hence,  $\{1, 3, 4, 6, 12\}$  is not a sublattice



Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$



→ Do we have any sub-lattice that is isomorphic to  $L_1^*$  or  $L_2^*$

① No sub set s.t. Hasse diagram is isomorphic to  $L_2^*$ , ∴ No sub-lattice is isomorphic to  $L_2^*$

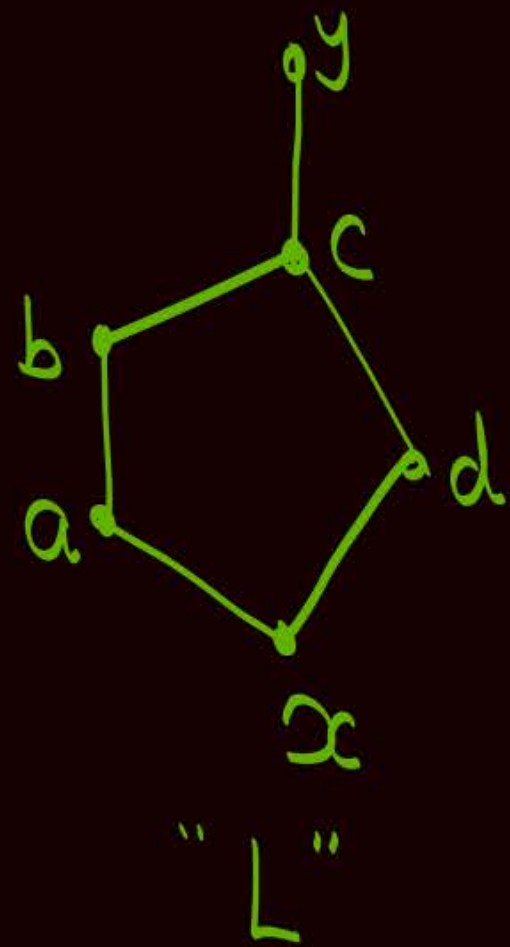
② We have two sub-sets  $\{1, 2, 3, 4, 12\}$  &  $\{1, 3, 4, 6, 12\}$  s.t. Hasse diagrams are isomorphic to  $L_1^*$  but none of them is a sub-lattice of given lattice i.e. No sub-lattice which is isomorphic to  $L_1^*$

No sub-lattice which is isomorphic to  $L_1^*$  or  $L_2^*$

Hence, given lattice is a distributive lattice



Q:- Check whether the following lattice is a distributive lattice or not?



(i) No. of elements  $> 4$ ,  $\therefore$  May or may not be distributive

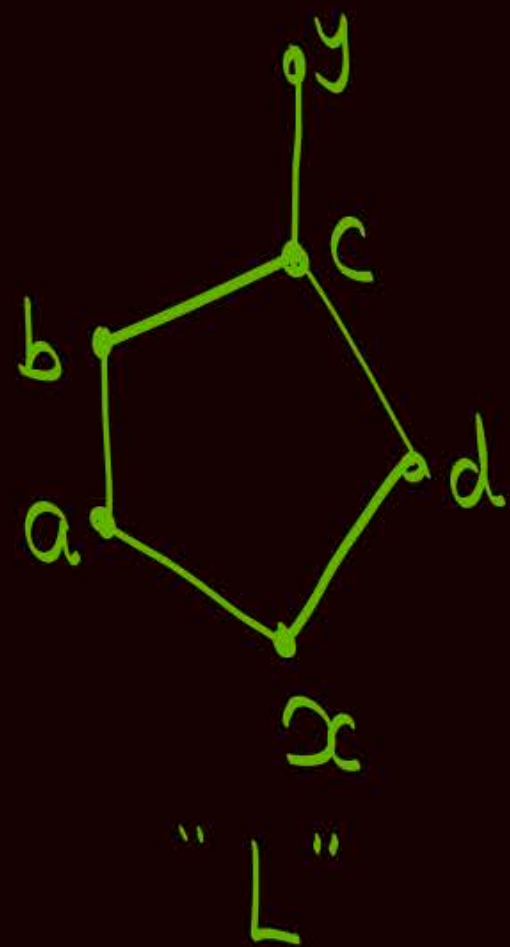
(ii) Not a TOSET,  $\therefore$  May or may not be distributive

(iii)  $\bar{x} = y$   
 $\bar{y} = x$   
 $\bar{a} = \text{DNE}$   
 $\bar{b} = \text{DNE}$   
 $\bar{c} = \text{DNE}$   
 $\bar{d} = \text{DNE}$

Every Element has at most '1' Complement,  
 $\therefore$  May or may not be distributive



Q: Check whether the following lattice is a distributive lattice or not?



$$\begin{aligned}\bar{x} &= y \\ \bar{y} &= x \\ \bar{a} &= \text{DNE} \\ \bar{b} &= \text{DNE} \\ \bar{c} &= \text{DNE} \\ \bar{d} &= \text{DNE}\end{aligned}$$

may or may not be distributive

But we have a sub-lattice  $\{x, a, b, c, d\}$  of lattice  $L$ , which is isomorphic to

$L_1^*$

$\therefore$  given lattice is not a distributive lattice

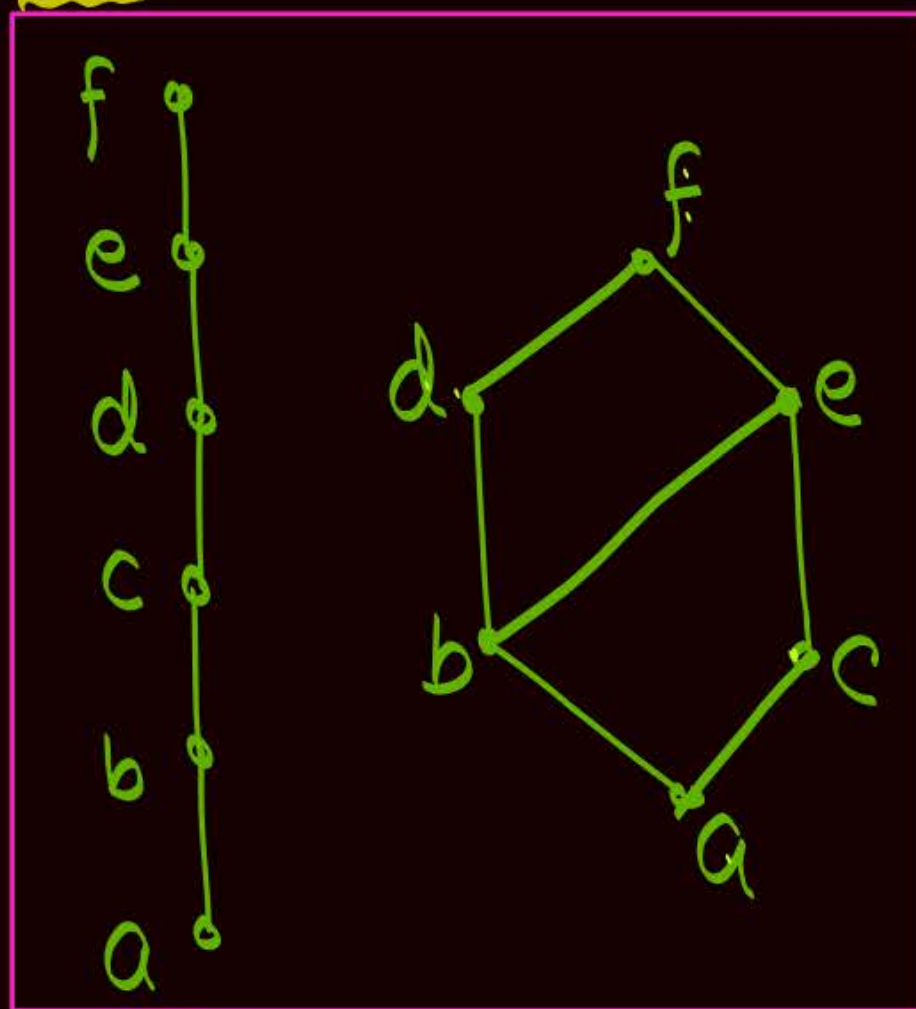
## \* Boolean Algebra / Boolean lattice

⑥ A lattice in which every element has Exactly one complement is called a Boolean lattice

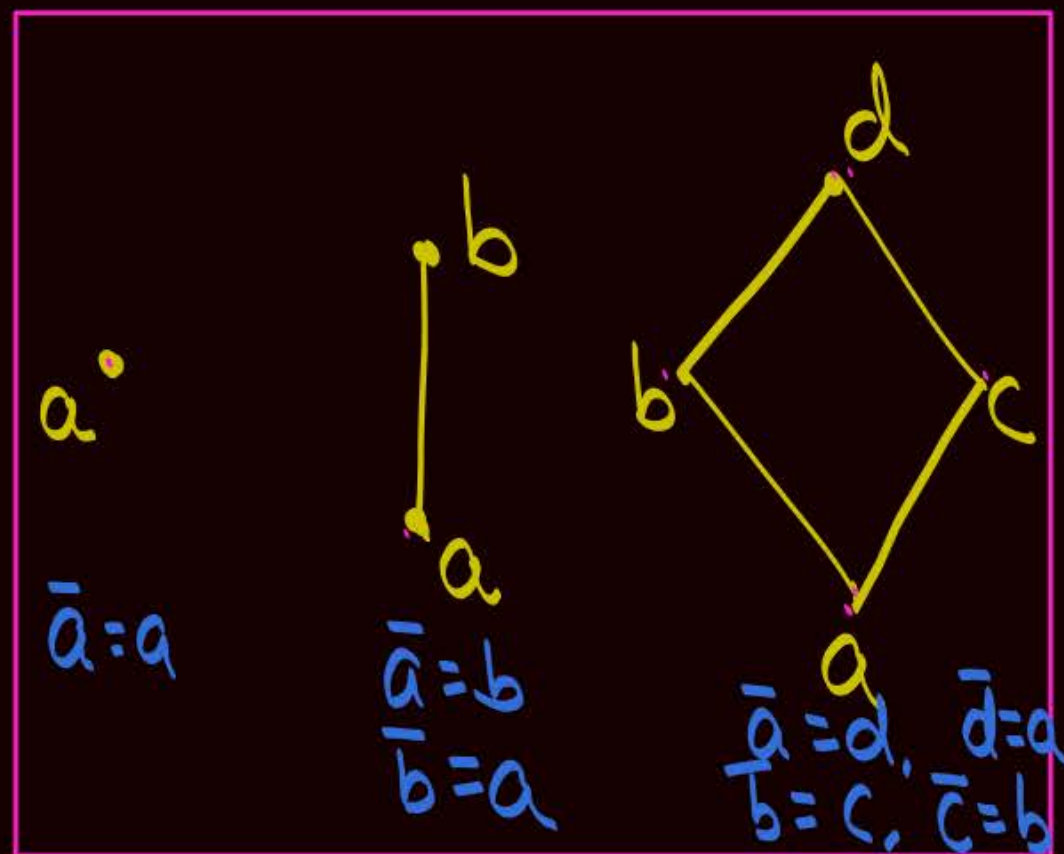
\* A lattice which is <sup>\*</sup>Complemented as well as distributive is called a Boolean lattice



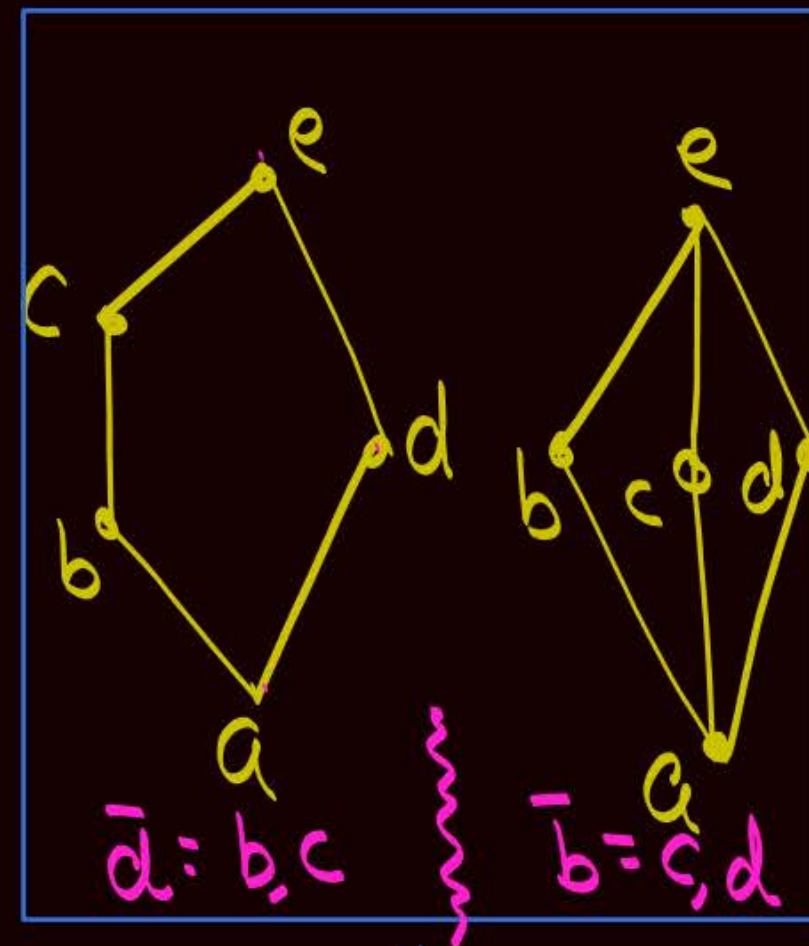
Note:-



Distributive,  
but not Complemented  
 $\Downarrow$   
 $\therefore$  Not a Boolean lattice



$\Downarrow$   
Complemented as well as  
distributive,  
 $\Downarrow$   
 $\therefore$  Boolean lattice



$\Downarrow$   
Complemented lattice but  
not a distributive lattice  
 $\Downarrow$   
 $\therefore$  Not a Boolean lattice

★ Note :- → If 'n' is positive integer such that  $D_n$  has no element which is a perfect square except 1, then 'n' is called a square free integer

Set of all positive divisors of 'n' →

eg let  $n=21$ ,

$$D_{21} = \{1, 3, 7, 21\}$$

No element is a perfect square (except 1)

∴ 21 is a square free integer

$$D_{20} = \{1, 2, 4, 5, 10, 20\}$$

it is a perfect sq. = ∴ 20 is not a square free integer



Note:

If  $n$  is a square free integer, then  $(D_n, \div)$  is a Boolean lattice,

and for an element  $x \in D_n$ ,

$$\overline{x} = \frac{n}{x}$$

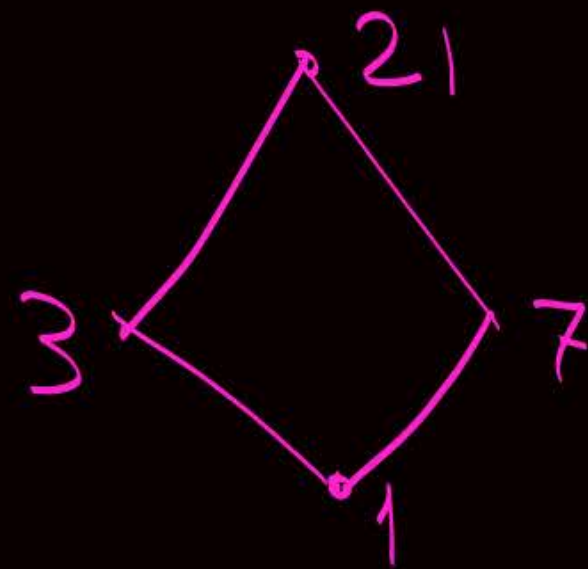
→ We know '21' is a square free integer

$$\& D_{21} = \{1, 3, 7, 21\}$$

then in Boolean lattice  $\{D_{21}, \div\}$

$$\overline{1} = \frac{21}{1} = 21, \quad \overline{7} = \frac{21}{7} = 3$$

$$\overline{3} = \frac{21}{3} = 7, \quad \overline{21} = \frac{21}{21} = 1$$



$$\Rightarrow \begin{aligned} \overline{1} &= 21 \\ \overline{21} &= 1 \\ \overline{3} &= 7 \\ \overline{7} &= 3 \end{aligned}$$

Note:-

① If lattice is a Boolean Algebra, then no. of Elements in lattice (Boolean Algebra) will always be in Power of '2'. {i.e.,  $2^0, 2^1, 2^2, \dots$ }  
but converse of the statement need not be true

"and"

\* If we have a Boolean algebra with  $2^n$  elements, then number of Edges in Hasse diagram corresponding to that Boolean Algebra will always be  $n \cdot 2^{n-1}$



# • Note → Boolean lattice

Lattice of  
size =  $2^0 = 1$

$a$

$$\bar{a} = a$$

Every element  
has Exactly '1'  
Complement,  
∴ Boolean lattice

$$0 \cdot 2^{0-1} = 0 \text{ edge}$$

Lattice of  
size =  $2^1 = 2$

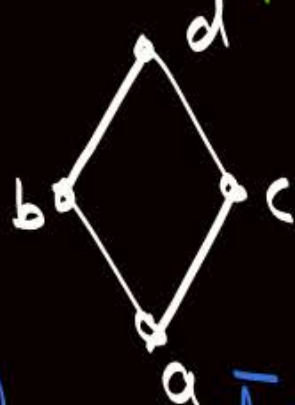


$$\bar{a} = b \text{ \& \; } \bar{b} = a$$

Every element  
has Exactly '1'  
Complement,  
∴ Boolean lattice

$$1 \cdot 2^{1-1} = 1 \text{ edge}$$

Lattice  
of size =  $2^2 = 4$



$$\bar{a} = d, \quad \bar{d} = a$$

$$\bar{b} = c, \quad \bar{c} = b$$

Every element  
has Exactly '1'  
Complement,  
∴ Boolean lattice

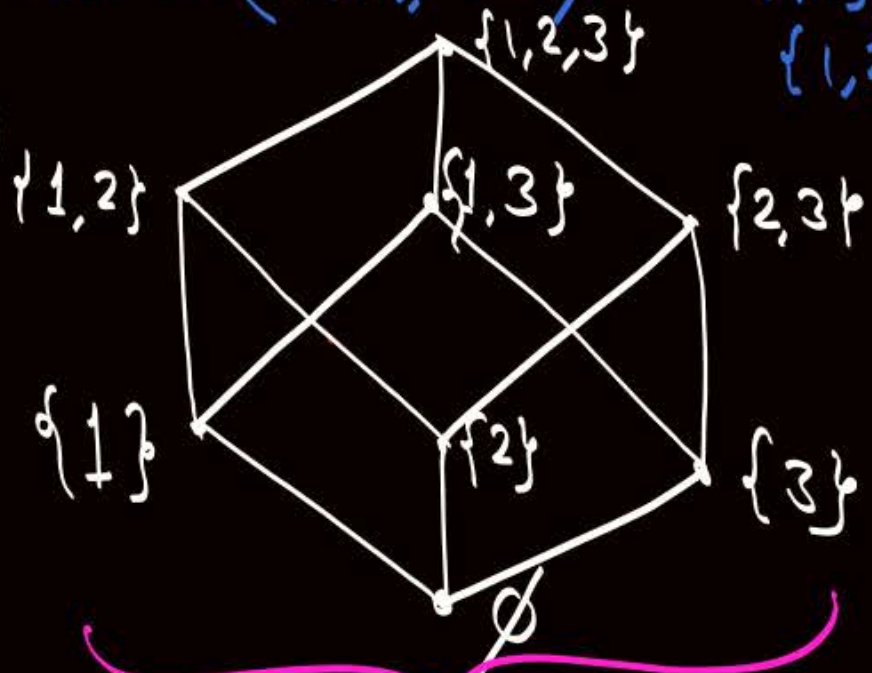
$$2 \cdot 2^{2-1} = 4 \text{ edge}$$



it is  
also a  
lattice  
of size = 4  
but it is  
not a  
Boolean  
lattice

Lattice of  
size =  $2^3 = 8$   
lattice  $(P(A), \subseteq)$

let  $A = \{1, 2, 3\}$   
 $P(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$



Unique Complement for  
Every element.

∴ Boolean lattice

$$3 \cdot 2^{3-1} = 3 \cdot 4 = 12 \text{ edges}$$

H.W.  
Q.1

Which of the following statements is/ are not true

- a) If  $A$  is any finite set then  $[P(A), \subseteq]$  is distributive lattice
- b) Every sub lattice of a distributive lattice is also a distributive lattice
- c) Every totally ordered set is a distributive lattice
- d) Every totally ordered set is bounded
- e) Every distributive lattice is bounded
- f) Every distributive lattice is a complemented lattice



H.W.

Which of the following lattice is /are not distributive?

- (A)  $(D_{125}, \div)$
- (B)  $(P(A), \subseteq)$
- (C)  $(D_{12}, \div)$
- (D)  $(\{1, 2, 3, 5, 30\}, \div)$
- (E)  $(\{1, 2, 3, 4, 12\}, \div)$

H.W.



Which of the following lattice is /are not Complemented ?

- (A)  $(D_{125}, \div)$
- (B)  $(P(A), \subseteq)$
- (C)  $(D_{12}, \div)$
- (D)  $(\{1, 2, 3, 5, 30\}, \div)$
- (E)  $(\{1, 2, 3, 4, 12\}, \div)$



H.W.

Which of the following lattice is /are boolean lattice?

- (A)  $(D_{125}, \div)$
- (B)  $(P(A), \subseteq)$
- (C)  $(D_{12}, \div)$
- (D)  $(\{1, 2, 3, 5, 30\}, \div)$
- (E)  $(\{1, 2, 3, 4, 12\}, \div)$



## 2 mins Summary



Topic

Complemented lattice

Topic

Sub-lattice

Topic

Distributive lattice

Topic

Boolean algebra



**THANK - YOU**